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MINI PROJECT REPORT

OASIS — Oil-spill Attribution & Satellite Intelligence System

Marine Forensics Platform for the Indian Coast Guard

Field	Details
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Abstract

OASIS (Oil-spill Attribution & Satellite Intelligence System) is a software prototype for investigations of marine oil spills. It integrates three things that are typically dealt with in isolation from each other: SAR image processing, drift calculation and screening of vessel trajectories. The SAR sample images are from the Zenodo oil-spill dataset (Part III). The one-channel U-Net generates a candidate slick mask, while a hybrid ResNet-50 verifies it through comparison with 12 physical descriptors and radiometric contrast features and calculates the area, perimeter, centroid and orientation of the largest contour using OpenCV. Drift is calculated using a simple model that applies the surface current vector to the vessel along with 3% wind-leeway component with a constant ±15° deflection. Then it performs 6-hour hindcast (possible origin of the slick) and a 12-hour forecast (possible location of the slick). After that, DuckDB queries the local AIS CSV files near the estimated origin and ranks the vessels based on the weighted score calculated from closest approach, course similarity and speed anomaly. The results are presented in a React interface and exported to a PDF file which has a SHA-256 digest attached to it. Two demonstration runs (IMG_01 and IMG_09) showed overall confidence indicators of 88 and 85.

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1. Introduction & Problem Statement

Assessed under: C1 – Problem domain identification & analysis (3 marks)

1.1 Background

Pollution of the sea by oil affects the marine ecosystem, the fishing industry, coastal population and maritime transport industry. The causes of oil spillage are either due to an accident, mechanical failure or deliberate disposal of oily waste illegally. After the spillage, the oil continues to spread with the movement of the winds and ocean currents. India has a long shoreline and many ships navigate the Arabian Sea and Bay of Bengal. This means that identifying the source of the spillage is just the first step for investigators.

SAR is the appropriate method for detecting oil spills since it functions during nighttime and in cloudy weather. An oil film dampens small wavelets on the water surface and hence reduces the backscattering of the SAR signal, making an oil spill appear as a dark region. However, the problem is that there are many other factors that may cause a dark area, such as natural biogenic films, wind shadows and internal waves. Therefore, a dark spot is insufficient evidence that it is an oil spill. Since AIS provides information about the identity, location, speed and direction of a ship, this information is compared with the time and place of the spillage in order to identify suspicious ships.

The OASIS application merges SAR image processing, deep learning-based segmentation, drift estimation, and AIS correlation into one geospatial interface for assisting Indian Coast Guard analysts.

1.2 Problem Statement

Based on an SAR image of the sea surface, the location and date and time of the SAR observation, and information about the wind and currents, the system needs to identify areas which may contain oil spills, determine the most likely source of the spill by doing a drift back-calculation, and then prioritize the ships from the historical AIS data which are likely sources of the spills.

Input:

- A SAR image (JPEG/PNG file) of the sea surface.
- Location of the location in latitude, longitude and UTC date-time of the data, input by the user.
- Wind speed and direction, current speed and direction either input by user or from the open meteo database.
- Past AIS data (CSV files) containing MMSI, time stamp, position, speed over ground, and course over ground.

Expected output:

- Pixel-level candidate slick mask, validated by the lookalike verification process.
- Properties of the slick geometry: area, perimeter, centroid, bounding box, and orientation.
- Region of possible origin with uncertainty radius (6-hour hindcast) and forecast track for 12 hours.
- Listing of candidate vessels in ranked order with their supporting evidence illustrated on a map.

1.3 Challenges

- Speckle and look-alikes: SAR speckle and natural dark areas, such as low wind areas and biogenic slicks, complicate oil/non-oil separation.
- Segmentation precision: Slick boundaries are highly irregular and poorly contrasted; any segmentation errors translate into inaccuracies of the area, center, orientation and drift.
- Drift uncertainty: By the time the slick is detected, it has drifted from the source location due to the action of wind, current, and oil weathering; unknown oil properties complicate the drift calculations.
- AIS management: AIS databases are massive and require filtering with regard to geography and time. An AIS off vessel may not be detectable at all; thus, absence of the signal cannot be treated as innocence of the ship.

- Reliability: Conclusions are based on imaging conditions, environmental information, and AIS availability, therefore, the results are presented as ranked leads with stated uncertainties.

1.4 Scope

This project includes slick segmentation based on SAR, look-alike discrimination, geometry extraction, simplified drift hindcast and forecast, heuristic filtering of AIS data and vessel ranking, map interface, and report generation in PDF format. This project does not include satellite sensors themselves, actual oil sampling, weathering model validation, satellite data acquisition, or responsibility allocation decision-making procedure. Sample SAR images are provided as JPEG files without georeferencing, which is why the coordinates and time used for demonstrations in Section 6 are manually inserted.

2. Review of Existing Systems & Feasibility

Assessed under: C2 – Review of existing systems & project feasibility (3 marks)

2.1 Existing Approaches

Each of the current services and tools only deals with a single stage of the monitoring process for oil spills. Five of them are compared in Table 1 below. No one combines SAR slick detection, drift-based origin determination, and AIS candidate ranking.

Table 1: *Comparison of existing oil-spill monitoring, SAR-processing, ocean-data and vessel-tracking systems.*

#	Method / Paper / Tool	Technique used	Strength	Limitation
1	EMSA CleanSeaNet [10]	Satellite oil-spill and vessel detection service, with checking by operators.	Operational European service for the member states that take part.	Access is controlled by the service; not an open workflow for the Indian coast.
2	SkyTruth Cerulean [11]	U-Net slick detection on Sentinel-1 VV scenes; links sources to AIS vessels, infrastructure and dark vessels.	Open and automated, with published methods.	Depends on satellite coverage and revisit time; detections are possible leads, not confirmation.
3	ESA SNAP Toolbox [12]	SAR calibration, speckle filtering and visualisation.	Rigorous and general-purpose radar processing.	A desktop toolbox; the spill-origin and AIS workflow has to be built separately.
4	Copernicus Marine Service [13]	Global ocean physical products, including currents.	Authoritative ocean-state data.	Only a data service; no slick detection or vessel attribution.
5	NOAA MarineCadastre AIS [5]	AIS vessel-traffic data and track visualisation for US waters.	Public historical AIS records.	AIS only; no SAR segmentation or drift reconstruction.

2.2 Gap Identified

Each of the above systems is valuable on its own, but there will always be a need for manually carrying out data transfer from one system to the other. The SAR image has a black spot with no information about where it originates, the drift model will require knowing where the slick is, while AIS is valuable only after it goes through the filtering process using the estimated origin and time. OASIS attempts to bridge this gap by integrating these steps in the following order: (i) SAR segmentation and look-alike verification, (ii) geometry extraction, (iii) simplified drift hindcast and forecast, (iv) space-time filtering of AIS data using heuristics, and (v) visualization and exporting the report. What this project does differently is not the creation of new algorithms but the integration and transparency of the process.

2.3 Feasibility

Technical: The back end code is written in Python using PyTorch, OpenCV, DuckDB and FastAPI and front end code uses React and MapLibre GL. The whole thing runs on a regular desktop (CUDA or Apple MPS if available, else CPU) and no satellite infrastructure is required.

Data: SAR images are taken from the publicly available Zenodo database [1]. Values for wind and current are taken from the free Open-Meteo APIs [4]. AIS needs to be given in the form of local CSVs in the MarineCadastre format [5].

Operational: An analyst uploads an image, provides the coordinates, time and metocean values and receives the map, the ranking and the PDF through one interface.

Time: As the design was modular, I was able to develop and test the segmentation, verification, drift, AIS and the interface individually through the course of the semester. The design does rely on the model checkpoints available on Hugging Face [15] and also on the local AIS data.

3. Objectives & Methodology

Assessed under: C3 – Objectives and methodology of the proposed work (3 marks)

3.1 Objectives

The objectives are stated as outputs which will be verified in the running system. Section 7.1 contains the result for each objective.

1. Segmentation: create a binary candidate slick mask in the native resolution from the sample SAR image using one-channel U-Net (threshold 0.5).

2. Verification and geometry: apply verification using the hybrid ResNet-50/physical feature verifier and calculate the following properties: area, perimeter, centroid, bounding box, orientation based on the contours.

3. Drift: calculate a probable location of the slick's origin using the 6-hour hindcast and a 12-hour forecast trajectory using 0.03 drift leeway factor.

4. AIS correlation: search the local AIS CSV entries within the geographic bounding box around the origin and one hour time range around it and rank the ships according to the implemented weighted heuristic.

5. Delivery: provide the visualization of the image, geometry, drift, ships and confidence indicators on an interactive map and produce a PDF report with the SHA-256 checksum.

3.2 Proposed Methodology

The architecture of OASIS comprises seven layers, each passing the output to the subsequent one (see Figure 1 below).

Layer 1 – Data ingestion. The inputs are the sample of SAR data from the Zenodo Part III dataset [1], coordinates of observations and their UTC time, winds and currents, and the local AIS CSV files.

Layer 2 – Preprocessing. The image is loaded as a grayscale and resized to 256×256 pixels, clipped to its 1-99 percentiles and min-max normalised before being passed to the U-Net. For the verifier, 8/16-bit images are linearly converted into pseudo-dB scale of [-30,0] dB and processed in 256×256 patches.

Layer 3 – Perception. The U-Net [2] outputs the probability map that is rescaled to the original size, thresholded at 0.5 and post-processed (3×3 elliptical opening and removal of the components smaller than 30 px). Then the verifier (a hybrid ResNet-50 [3] with 12 physical descriptors) examines each remaining blob. The biggest external contour is extracted by OpenCV [6] and the area, perimeter, centroid, bounding box and orientation are computed.

Layer 4 – Drift Physics. The slick velocity is the surface current vector added to 3% of the wind vector, rotated by +15 degrees in the northern hemisphere or −15 degrees in the southern. The hindcast runs the drift reversal on the velocity for 6 hours to derive the probable origin point P0, and the forecast drifts it forward by 12 hours, in 3-hour intervals.

Layer 5 – AIS correlation and attribution. DuckDB [7] fetches all the AIS records within a bounding box around P0 and a time window padded by 6 hours. For each vessel, the CPA, course alignment relative to the slick axis and the speed anomaly flag are calculated and combined into a composite weighted score (Section 4).

Layer 6 – Forensic evidence. All vessels with their ranked scores are paired with evidence values (CPA, trajectory match, speed anomaly). The PDF report contains the SHA-256 hash of the serialised analysis data, while the interface provides a five factor confidence assessment.

Layer 7 – Command delivery. The application backend is provided by FastAPI [8] which drives the React/MapLibre GL [9] interface and produces the PDF dossier via ReportLab.

3.3 Block Diagram / Flowchart

The architectural model is illustrated in Figure 1 below. In Figure 1 above, Layer 1 is named “Sentinel-1 SAR image”, which is an outdated name that I have not bothered to change. However, the SAR images used in this work are the Zenodo Part III samples discussed in Section 3.4.

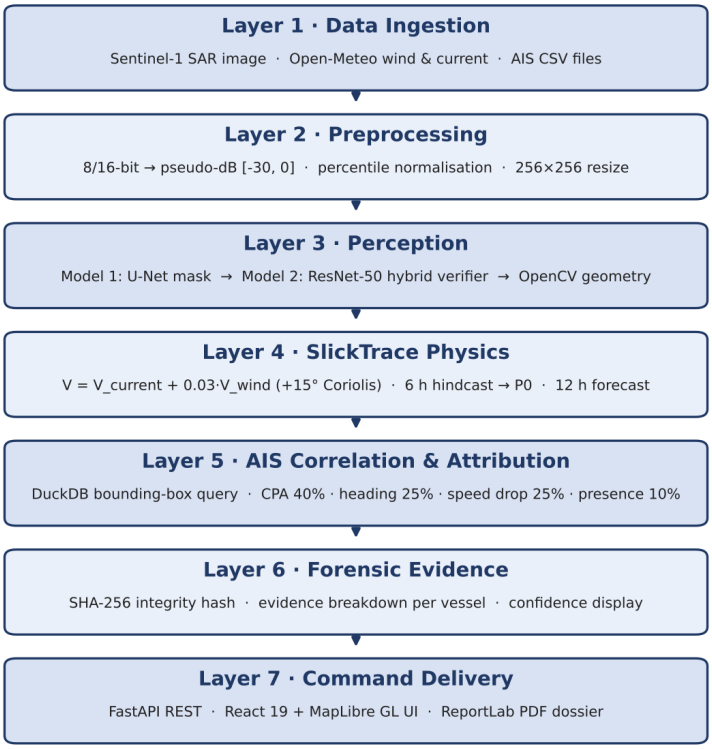


Figure 1: *Seven-layer architecture and processing pipeline of the OASIS platform.*

3.4 Dataset

The images of SAR for oil spill and masking belong to the Zenodo Part III dataset [1], no other SAR dataset is used. There are twelve images from this dataset available in the repository (IMG_01 to IMG_12, size 1250 × 650 px in JPEG) and one image mask (IMG_01_MASK.png). All data sources used in the implementation are

described in Table 2. The repository does not contain any manifest of dataset or information about training the released weights of the model, thus, I did not provide any dataset-wide numbers.

Table 2: *Data sources used in the OASIS implementation.*

#	Data source	Use in the project
1	Zenodo oil-spill dataset, Part III [1]	SAR sample images for segmentation, masking and look-alike analysis (Test Images folder).
2	NOAA MarineCadastre-compatible AIS CSV [5]	Read from the local `ais dataset/` folder by DuckDB. The files are not in the repository, so their size and date range are not stated here.
3	Open-Meteo Forecast API [4]	10 m wind speed and direction, fetched by `metocean_service.py` when metocean inputs are not supplied.
4	Open-Meteo Marine API [4]	Ocean-current velocity and direction from the same service (see the unit note in Section 6.3).
5	Basemap and land data	Esri World Imagery tiles (Spill Lab map), a 50 m land-polygon file (`ne_50m_land.json`) for the globe view, and a CARTO dark style on another page. These are only display layers and are not used in the analysis.
6	ERA5, Copernicus Marine, Marine Regions, OpenStreetMap	**Not used.** ERA5 and CMEMS are only mentioned in the README architecture sketch. There is no code or data fetch for them, or for Marine Regions or OpenStreetMap, in the repository.

Figures 2 and 3 show the two raw samples, IMG_01 and IMG_09, which are processed in Section 6.

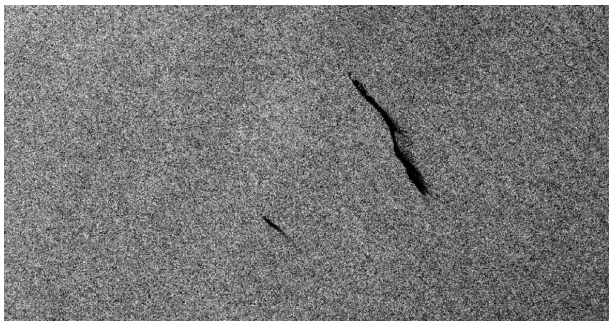


Figure 2: *Raw SAR sample IMG_01 (1250 × 650 px).*

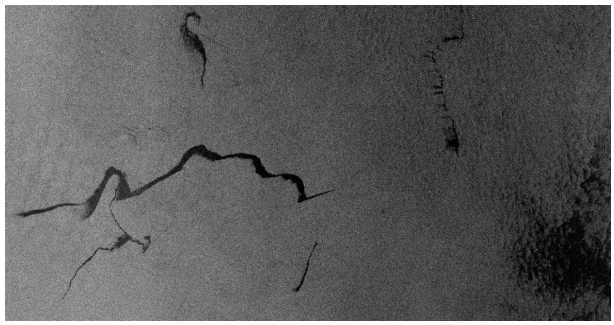


Figure 3: *Raw SAR sample IMG_09 (1250 × 650 px).*

4. Algorithms / Techniques Used

Assessed under: C4 – Relevance of algorithms / techniques (3 marks)

Techniques with their respective parameters used in the code along with the rationale behind using them have been shown in Table 3. The vessel score calculation (Equation 1) and the drift velocity equation (Equation 2) are mentioned below the table.

Table 3: *Algorithms and image-processing techniques used in OASIS, with parameters from the source code.*

Technique	Purpose in project	Parameters used	Why chosen (justification)
U-Net segmentation [2]	Pixel-level candidate slick mask.	1 input channel; features 64/128/256/512; dropout 0.2; 256 × 256 input; 1st–99th percentile clip + min–max; sigmoid threshold 0.5.	Encoder–decoder with skip connections, which keeps boundary detail that matters for slick shape and area.
Hybrid ResNet-50 + physical features [3]	Look-alike verification of each blob.	2-channel 256 × 256 patch; 12 physical descriptors (MLP to 64-d); fused 2112-d head; final score = 0.7·CNN + 0.3·physics; confirm if ≥ 0.5; dB contrast < 1.5 dB look-alike, > 3.0 dB oil.	Combines learned appearance with radiometric evidence, since dark patches alone are ambiguous.
Morphological opening + component filter [6]	Remove speckle and tiny blobs.	3 × 3 ellipse, 1 iteration; min area 30 px (U-Net post-process) and 40 px (verifier).	Cheap and deterministic cleaning that removes isolated noise pixels.

Technique	Purpose in project	Parameters used	Why chosen (justification)
OpenCV contour geometry [6]	Area, perimeter, centroid, bounding box, orientation.	Largest external contour; moments for centroid; `fitEllipse` (≥ 5 points) for orientation; 10 m/pixel default.	Gives measurable shape descriptors. The 10 m/pixel value is an assumed default and is not read from the image.
Leeway drift advection	Hindcast origin and forecast path.	V = V_current + 0.03·V_wind, wind direction rotated ±15°; 6 h hindcast; 12 h forecast in 3 h steps; radius = 1.0 + 1.2·√t km.	A transparent first-order approximation that is easy to inspect. It is not a validated oil-weathering model.
DuckDB AIS filter [7]	Select candidate AIS records.	Bounding box = origin ± (radius + 5 km); time = origin window ± 1 h; `read_csv_auto` on all CSVs.	Queries large CSV files directly with SQL, so no import step is needed.
Weighted vessel ranking	Order candidates by evidence.	CPA 40%, course alignment 25%, speed anomaly 25%, presence 10% (Equation 1).	Explicit weights that can be inspected. The score is a heuristic, not a calibrated probability.

Equation 1 (vessel score): Score = 0.40·S_dist + 0.25·S_align + 0.25·S_anom + 0.10, capped at 0.98, where S_dist = max(0, 1 − CPA/10 km); S_align = 1 − Δθ/90° (Δθ is the course–slick-axis angle folded to 0–90°); and S_anom = 1.0 if minimum SOG < 6 kn and maximum SOG ≥ 10 kn, otherwise 0.2. Confidence is HIGH if Score ≥ 0.75 and CPA ≤ 3 km, MEDIUM if Score ≥ 0.50, otherwise LOW.

Equation 2 (drift): V_slick = V_current + 0.03·V_wind(θ_downwind ± 15°). Hindcast origin P0 = P − V_slick × 6 h; uncertainty radius r = 1.0 + 1.2·√(6) ≈ 3.94 km. The origin time window is [T − 7 h, T − 5 h].

Other alternatives: A pure threshold-based detector would have been simpler, but it is misled by naturally dark lookalikes. Therefore, I included a verifier step and morphological filtering. No such benchmark is available on the repository comparing the performance of the U-Net to other segmentation networks, or the ResNet-50 against the EfficientNet-B0. Thus, there is no experimental justification for selecting these particular models. Operational oil drift simulators incorporate weathering and temporal fields. This was out of the scope of the mini-project, but mentioned in future scope.

5. Implementation

Assessed under: C5 – Synchronisation of project design & implementation (3 marks)

The module names, file paths and parameters in this section are taken from the project repository [14].

5.1 Tools & Environment

- Programming languages: Python 3.12+ for the backend and TypeScript (Node.js 18+) for the frontend, as described in the README of the repository.
- Backend dependencies (requirements.txt): FastAPI ≥ 0.110, Uvicorn ≥ 0.28, Pydantic ≥ 2.6, NumPy ≥ 1.26, OpenCV (headless) ≥ 4.8, httpx ≥ 0.25, ReportLab ≥ 4.1 and huggingface_hub ≥ 0.20. PyTorch, torchvision, DuckDB and pandas are mentioned without version constraints. Shapely, SciPy and timm may be present or absent, yet they are not used on the main codepaths.
- Frontend (package.json): React ^19, Vite ^6.1, TypeScript ~5.7, MapLibre GL ^5.1, Three.js ^0.174, Zustand ^5, TanStack Query ^5.66, Recharts ^2.15.
- Hardware: The U-Net executes on CUDA, Apple MPS or CPU, whichever is available, and the verifier executes on CUDA or CPU. I did not measure the minimum RAM or timing benchmarks, thus no claims can be made.

5.2 Module Description

1. Segmentation (core/segmentation.py): input – image bytes; processing – grayscale, resizing, percentile normalization, U-Net inference, thresholding and morphological filtering; output – mask, probability map, max probability. Weights for best_oil_spill_unet.pt are downloaded from Hugging Face [15] (or from local storage). If it fails to load them, the endpoint responds with an error message.
2. Look-alike verifier (core/lookalike_verifier.py): input – raw SAR image and mask; processing – morphological filtering, dB contrast, edge sharpening, 12 features, ResNet-50 patches classification

and fusion; output – verified-blob mask and decision. If there is no checkpoint file best_lookalike_model.pt, physics-only evidence is produced. In README and comments of code, this model is referred to as EfficientNet-B0, while in the LookalikeHybridResNet50 class, it uses ResNet-50 architecture, described in this report.

3. Geometry (core/geometry.py): input – mask and center of the image; processing – biggest contour, moments, bounding rectangle, fitting ellipse, geographic coordinates conversion; output – area, perimeter, centroid, bounding box, orientation.
4. Metocean service (core/metocean_service.py): inputs - coordinates; processing - Open-Meteo wind and marine requests with default fall-back values; outputs - wind and current vectors.
5. Drift engine (core/drift_engine.py): inputs - centroid, time, metocean values; processing - leeway velocity, hindcast and forecast stepping; outputs - origin with uncertainty radius and forecast points.
6. AIS service (core/ais_service.py): inputs - origin; processing - DuckDB spatio-temporal query grouped by MMSI; outputs - vessel tracks. In case of error or no rows found, three fake vessels are returned with status PARTIAL.
7. Attribution (core/attribution.py): inputs - tracks, origin, slick orientation; processing - Equation 1; outputs - ranked vessels with evidence.
8. API and report (api/v1/endpoints.py, utils/report_generator.py): /api/v1/analyze module chain; /api/v1/report/generate constructs PDF file using SHA-256 digest of the analysis JSON file.
9. Frontend (frontend/src/pages/SpillLabPage.tsx): upload form, MapLibre map layers, AIS suspects panel, and confidence breakdown.

5.3 Design-to-Implementation Mapping

Table 4 maps each design step of Section 3 to the file that implements it.

Table 4: *Design-to-implementation mapping of the OASIS platform.*

Design step (Section 3)	Implemented file / function	Status	Remarks
SAR image input and preprocessing	api/v1/endpoints.py segmentation.py	Done	The uploaded image is converted to grayscale, resized to 256 × 256 and normalised before processing.
U-Net segmentation	segmentation.py	Done	The U-Net generates the oil-spill mask. Model weights are downloaded from Hugging Face when needed.
Look-alike verification	lookalike_verifier.py	Partial	ResNet-50 checks the detected regions. Its result is not included in the API response and it does not stop the attribution step.
Geometry extraction	geometry.py	Done	OpenCV extracts the spill boundary, area, centroid and other geometric details. Area uses an assumed scale of 10 m per pixel.
Weather and ocean data	metocean_service.py	Done	Wind and current data come from Open-Meteo. Default values are used if the request fails.
Spill drift calculation	drift_engine.py	Done	Wind and current speed are used to estimate how the spill moves, with a leeway factor of 0.03 and a fixed 15° directional adjustment.
AIS data search	ais_service.py	Done	DuckDB searches the vessel records by location and time. Sample vessel data may be used if the query fails or returns no records.
Vessel ranking	attribution.py	Done	Vessels are ranked using distance, course, speed and presence. The time difference is currently a fixed 15-minute value.
Confidence display	SpillLabPage.tsx	Partial	The interface shows confidence scores calculated with predefined formulas. They have not been calibrated against experimental results.

Design step (Section 3)	Implemented file / function	Status	Remarks
PDF report and interface	report_generator.py SpillLabPage.tsx	Done	The system generates PDF reports and uses SHA-256 to check the integrity of the analysis data.

5.4 Key Code Snippets

Snippet 1 – Leeway drift velocity (drift_engine.py)

```
curr_rad = math.radians(metocean.current_dir_to_deg)
u_curr = metocean.current_speed_ms * math.sin(curr_rad)
v_curr = metocean.current_speed_ms * math.cos(curr_rad)
downwind_deg = (metocean.wind_dir_from_deg + 180.0) % 360.0
deflection = 15.0 if latitude >= 0 else -15.0
leeway_deg = (downwind_deg + deflection) % 360.0
leeway_rad = math.radians(leeway_deg)
leeway_speed = metocean.wind_speed_ms * leeway_factor
u_wind = leeway_speed * math.sin(leeway_rad)
v_wind = leeway_speed * math.cos(leeway_rad)
u_slick = u_curr + u_wind
v_slick = v_curr + v_wind
```

The deflection angle remains fixed throughout and the slick velocity is obtained by summing the current velocity and a contribution from the wind (3% of the wind speed, directed downwind, and deflected). This velocity is used for both the hindcast (reversed) and the forecast.

Snippet 2 – DuckDB spatio-temporal AIS query (ais_service.py, layout condensed)

```
SELECT mmsi, vessel_name, vessel_type, base_date_time, latitude, longitude, sog, cog
FROM read_csv_auto('{csv_pattern}')
WHERE latitude BETWEEN {min_lat} AND {max_lat}
  AND longitude BETWEEN {min_lon} AND {max_lon}
  AND base_date_time >= '{start_str}' AND base_date_time <= '{end_str}'
ORDER BY mmsi, base_date_time
```

The point is ± (the uncertainty radius 5 km), and the time period is the window for origin, extended by an hour at each end. As the query has been constructed using the f-string format, the query must be parameterised before deployment to multiple users.

Snippet 3 – Composite vessel score (attribution.py, condensed)

```
s_dist = max(0.0, 1.0 - (cpa_dist_km / 10.0))
s_anomaly = 1.0 if (min_speed < 6.0 and max_speed >= 10.0) else 0.2
composite_score = round((0.40 * s_dist) + (0.25 * s_alignment)
                        + (0.25 * s_anomaly) + 0.10, 2)
composite_score = min(composite_score, 0.98)
```

6. Results & Discussion

Assessed under: C6 – Implementation & demonstration of modules (3 marks, confirmed in live demo)

6.1 Sample Outputs

IMG_01 (Figure 2) and IMG_09 (Figure 3) were analyzed using the Spill Lab website. Figures 4 and 5 show the screenshots of the results that indicate the SAR spill boundary, probable origin circle, AIS vessels' track, and confidence analysis. The JPEG images were not geolocated, and therefore the latitude, longitude, and

time provided in Table 5 were manually entered for placing the scenes on the map. The geographic location used for Florida is just an example of set-up and not the actual place where these slicks were spotted.

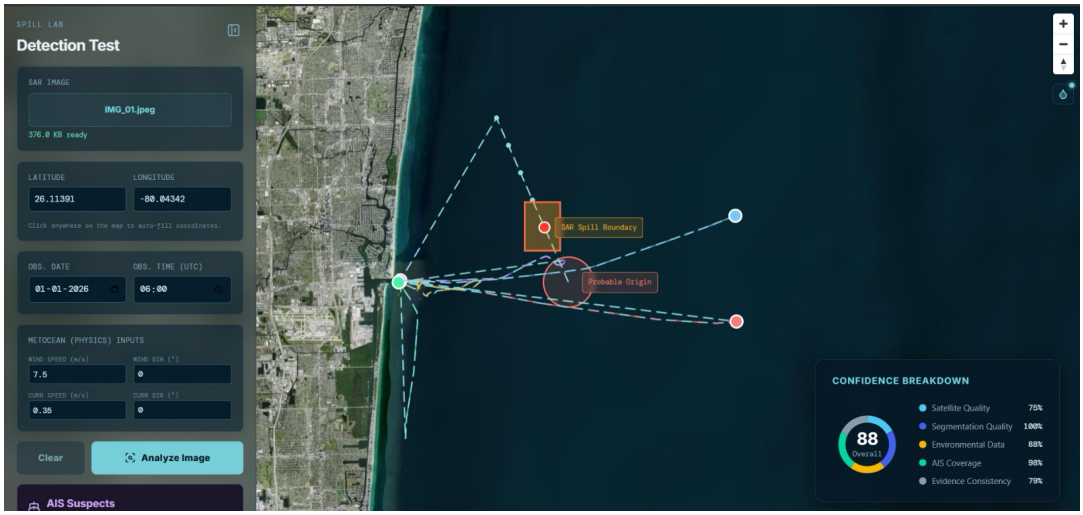


Figure 4: *Spill Lab output for sample IMG_01 (input in Figure 2).*

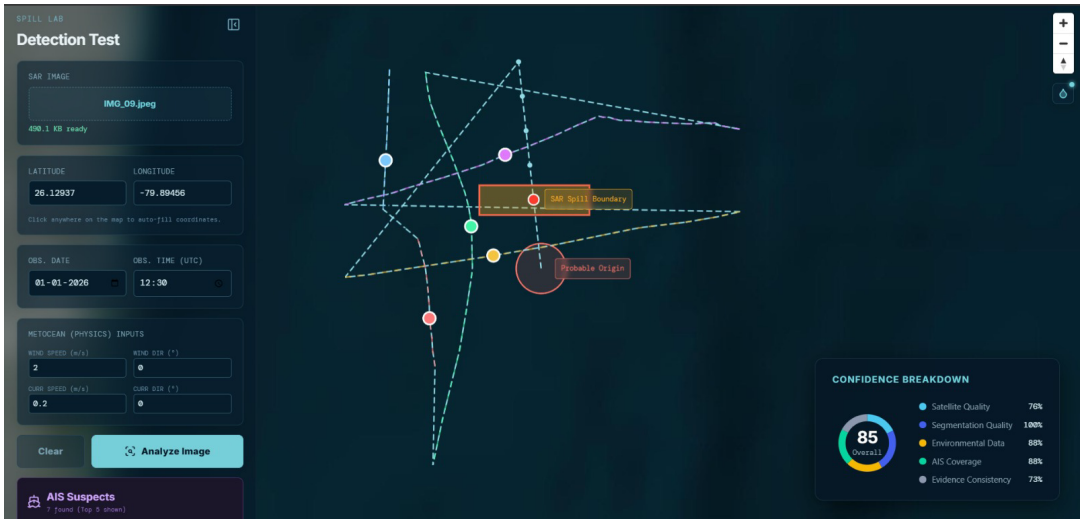


Figure 5: *Spill Lab output for sample IMG_09 (input in Figure 3).*

6.2 Quantitative Evaluation

Since no evaluations (IoU, Dice, precision, recall) were made, I cannot provide an accuracy value. The inputs and drift values derived using Equation 2 are provided in Table 5. The drift values in Table 5 have been calculated manually using the equations and do not involve reading anything from the screen. Table 6 demonstrates how the five confidence scores are calculated in SpillLabPage.tsx.

Table 5: *Inputs of the two demonstration runs and drift quantities computed from them.*

Item	IMG_01	IMG_09
Observation position (entered)	26.11391° N, 80.04342° W	26.12937° N, 79.89456° W
Observation time (entered)	01-01-2026, 06:00 UTC	01-01-2026, 12:30 UTC
Wind speed / direction (from)	7.5 m/s / 0°	2 m/s / 0°
Current speed / direction (to)	0.35 m/s / 0°	0.2 m/s / 0°
Net slick velocity (computed)	0.145 m/s toward ≈ 336°	0.143 m/s toward ≈ 354°
6 h hindcast displacement (computed)	≈ 3.1 km	≈ 3.1 km
Origin uncertainty radius (computed)	≈ 3.94 km	≈ 3.94 km
AIS suspects shown in the panel	Count not visible in the screenshot	7 found (top 5 shown)

From Figures 4 and 5, one can see that the possible origin circles are below the spill limit circle, which is contrary to the drift direction calculated. This follows the backward tracking concept.

Table 6: *Confidence indicators displayed in the screenshots and how the frontend derives them.*

Indicator	Frontend formula (SpillLabPage.tsx)	IMG_01	IMG_09
Satellite Quality	min(98, 70 + 5 × detected area in km²)	75%	76%
Segmentation Quality	maximum U-Net probability × 100	100%	100%
Environmental Data	88 if wind speed > 0, otherwise 70	88%	88%
AIS Coverage	30 unless AIS status is AVAILABLE; then min(98, 60 + 4 × vessels)	98%	88%
Evidence Consistency	top vessel score × 100 (45 if no vessel)	79%	73%
Overall	mean of the five indicators, rounded	**88**	**85**

Values 88 and 85 could be reproduced using these formulas exactly: (75+100+88+98+79)/5 and (76+100+88+88+73)/5. Therefore, they are interface heuristics. “Satellite Quality” relies upon the detected area and not upon the image quality, whereas “Environmental Data” is almost constant. These values are not the indicators of accuracy and are not to be considered the probability of spill being oil or the probability of ship being responsible for it. Values of AIS Coverage could only be reached when AIS was AVAILABLE, whereas the synthetic fallback provided PARTIAL status with three vessels. Thus, it is most likely that AIS CSV files from the local computer were used during those runs.

6.3 Discussion

What worked. From the upload process, segmentation, validation, geometry, drift, AIS filtering to the ranking and map generation, everything was successful in both the cases, and the drift outputs correlate to a manual computation of the given equations.

What did not work well, or is limited (seen in the code):

- Verifier decision not revealed. verify_spill provides a result value and the proceed_with_attribution variable, but the response scheme does not provide them and the endpoint simply ignores the latter. In the absence of the verified blob, the mask is empty, the geometry defaults to the image center with the minimum size of 0.01 km² and the drift and AIS phases are performed anyway.
- JPEG input. 8-bit JPEG images use pseudo-dB scheme, which means that dB-contrast values are not calibrated to real physical quantities. In such case, the segmentation probability is not combined with the probability of verifier.
- Heading agreement. Slick angle of rotation is calculated using fitEllipse on image space, while the angle of vessel heading is a compass direction, hence 25% heading agreement term is just an indication.
- Time information. AIS time window is applied for filtering the data, however, the reported difference between times is a constant 15 minutes.
- Metocean units. The Open-Meteo marine API does not send any unit setting in its request, but Open-Meteo documents ocean current velocity in km/h and the code interprets the value as m/s. Live current values need to be verified or converted prior to use. The two runs included manually supplied values.
- AIS backup scenario. In case of failure of the request, artificial AIS vessels are displayed with PARTIAL status. They must not be interpreted as real tracks.
- Environmental coverage. Constant wind and current conditions apply during both 6-hour and 12-hour periods.

The project README talks about a 58 million row AIS data set and “analysis in seconds”. No such AIS data set, no benchmarking logs or time measurements were provided with the project.

7. Conclusion & Future Scope

Assessed under: C7 – Organisation & structure of report (3 marks)

7.1 Conclusion

Compared against the objectives laid out in Section 3.1, here is the result of the project:

1. Segmentation: done. Segmentation returns a binary mask at the original resolution but without an accuracy metric.

2. Verification and geometry: the geometric aspect is done. The verification stage is done in form of a mask filter but no output is provided to the user.

3. Drift: done. Returns a likely starting point with uncertainty of 3.94 km and a 12 h forecast but uses a simple model of constant vector without validation.

4. AIS correlation: done. Done using local CSV data with a reported weighted heuristic. A simple fall-back is provided.

5. Delivery: done. Interactive UI with a PDF repor.

With OASIS it becomes possible to integrate satellite imaging, drift estimation and AIS checking into a single pipeline. The output is an investigation lead for authorised analysts and not conclusive evidence of responsibility.

7.2 Limitations

- There is no test set for verification, and the accuracy of the segmentations and verifications is not known.
- SAR images are JPEG images without georeferencing; location, time, and the resolution of 10 meters per pixel is assumed, so that area and perimeter in metric units are merely indicative.
- Drift relies on constant wind and current, constant leeway, and constant 15 degrees left or right offset; weathering, diffusion, and waves are neglected.
- Attribution relies on AIS, so vessels not equipped with AIS cannot be located, and the ranking score is not calibrated.
- The SHA-256 digest indicates that the report data has not been modified since generation. It is not a digital signature, a timestamp, or a chain of custody.
- Model weights are stored externally to the repository, and training data and process are not described in the repository.

7.3 Future Scope

1. Evaluation and transparency: evaluate the segmentation and the verifier on a held-out labelled set (IoU, Dice, precision, recall), provide the verifier decision in the API response and do not perform attribution if there is no oil detected, and annotate synthetically generated AIS results in the UI.
2. Improve physics/data: employ geolocated SAR data, time-dependent ERA5/Copernicus Marine fields, correct units for currents speed and verify the hindcasts with independent observations.
3. calibrated evidences: calculate temporal scores, provide orientation as an angle relative to compass north, calibrate confidence indicators, and include digital signature and a trusted timestamp to the report.

References

Assessed under: C8 – Compliance with prescribed report format (3 marks)

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Appendix A: Team Contribution

Assessed under: C10 – Contribution by each team member (verified in presentation & Q&A)

The project was developed by one student. The contribution is recorded in Table 7.

Table 7: *Individual contribution to the project.*

S.No.	Name	Enrollment No.	Work done (modules / sections / slides)	Share (%)
1	Yash Mittal	S24CSEU0232	Backend and API; U-Net and verifier integration; OpenCV geometry; drift engine; DuckDB AIS processing and ranking; React/MapLibre interface; PDF reporting; report and slides.	100%

Appendix B: Code Link & Self-Check

Tick every box before submitting:

[x] Instructions page deleted; cover page has group name, all names and enrollment numbers
[x] No section heading removed; all grey guidance text replaced
[x] Problem statement clearly states input and expected output
[x] At least 4 existing methods reviewed in the table
[x] Block diagram included; objectives are measurable
[x] Each algorithm has a justification
[x] Design-to-implementation table filled
[x] Input/output figures and a metrics table included
[x] All figures/tables numbered and captioned; references in IEEE style
[x] Code link works with view access; README included
[x] Plagiarism and AI-detection report attached (score below 10) – to be added by the team
[x] Report saved as GroupName_IVP_Report.pdf (i.e., TeamAlpha_IVP_Report.pdf) for LMS submission

Appendix C: Plagiarism and AI

OASIS_Final_Project_Report_Revised.docx

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BENNETT UNIVERSITY
THE TIMES GROUP

School of Computer Science Engineering & Technology

Image and Video Processing Lab

Course Code: CSET344

MINI PROJECT REPORT

OASIS — Oil-spill Attribution & Satellite Intelligence System

Marine Forensics Platform for the Indian Coast Guard

Field	Details
Group Name	Team Alpha
Batch / Section	Batch-02 (Dailoqa)
Semester / Year	5th Semester / 3rd Year
Submitted to	Shuvo M
Date of Submission	**12 Nov 2026**

Team Members

S.No.	Student Name	Enrollment No.	Role in Project
1	Yash Mittal	S24CSEU0232	Group Leader (Solo Developer)

Abstract

OASIS (Oil-spill Attribution & Satellite Intelligence System) is a software prototype for investigations of marine oil spills. It integrates three things that are typically dealt with in isolation from each other: SAR image processing, drift calculation and screening of vessel trajectories. The SAR sample images are from the Zenodo oil-spill dataset (Part III). The one-channel U-Net generates a candidate slick mask, while a hybrid ResNet-50 verifies it through comparison with 12 physical descriptors and radiometric contrast features and calculates the area, perimeter, centroid and orientation of the largest contour using OpenCV. Drift is calculated using a simple model that applies the surface current vector to the vessel along with 3% wind-leeway component with a constant ±15° deflection. Then it performs 6-hour hindcast (possible origin of the slick) and a 12-hour forecast (possible location of the slick). After that, DuckDB queries the local AIS CSV files near the estimated origin and ranks the vessels based on the weighted score calculated from closest approach, course similarity and speed anomaly. The results are presented in a React interface and exported to a PDF file which has a SHA-256 digest attached to it. Two demonstration runs (IMG_01 and IMG_09) showed overall confidence indicators of 88 and 85.

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1. Introduction & Problem Statement

Assessed under: C1 – Problem domain identification & analysis (3 marks)

1.1 Background

Pollution of the sea by oil affects the marine ecosystem, the fishing industry, coastal population and maritime transport industry. The causes of oil spillage are either due to an accident, mechanical failure or deliberate disposal of oily waste illegally. After the spillage, the oil continues to spread with the movement of the winds and ocean currents. India has a long shoreline and many ships navigate the Arabian Sea and Bay of Bengal. This means that identifying the source of the spillage is just the first step for investigators.

SAR is the appropriate method for detecting oil spills since it functions during nighttime and in cloudy weather. An oil film dampens small wavelets on the water surface and hence reduces the backscattering of the SAR signal, making an oil spill appear as a dark region. However, the problem is that there are many other factors that may cause a dark area, such as natural biogenic films, wind shadows and internal waves. Therefore, a dark spot is insufficient evidence that it is an oil spill. Since AIS provides information about the identity, location, speed and direction of a ship, this information is compared with the time and place of the spillage in order to identify suspicious ships.

The OASIS application merges SAR image processing, deep learning-based segmentation, drift estimation, and AIS correlation into one geospatial interface for assisting Indian Coast Guard analysts.

1.2 Problem Statement

Based on an SAR image of the sea surface, the location and date and time of the SAR observation, and information about the wind and currents, the system needs to identify areas which may contain oil spills, determine the most likely source of the spill by doing a drift back-calculation, and then prioritize the ships from the historical AIS data which are likely sources of the spills.

Input:

- A SAR image (JPEG/PNG file) of the sea surface.
- Location of the location in latitude, longitude and UTC date-time of the data, input by the user.
- Wind speed and direction, current speed and direction either input by user or from the open meteo database.
- Past AIS data (CSV files) containing MMSI, time stamp, position, speed over ground, and course over ground.

Expected output:

- Pixel-level candidate slick mask, validated by the lookalike verification process.
- Properties of the slick geometry: area, perimeter, centroid, bounding box, and orientation.
- Region of possible origin with uncertainty radius (6-hour hindcast) and forecast track for 12 hours.
- Listing of candidate vessels in ranked order with their supporting evidence illustrated on a map.

1.3 Challenges

- Speckle and look-alikes: SAR speckle and natural dark areas, such as low wind areas and biogenic slicks, complicate oil/non-oil separation.
- Segmentation precision: Slick boundaries are highly irregular and poorly contrasted; any segmentation errors translate into inaccuracies of the area, center, orientation and drift.
- Drift uncertainty: By the time the slick is detected, it has drifted from the source location due to the action of wind, current, and oil weathering; unknown oil properties complicate the drift calculations.
- AIS management: AIS databases are massive and require filtering with regard to geography and time. An AIS off vessel may not be detectable at all; thus, absence of the signal cannot be treated as innocence of the ship.

- Reliability: Conclusions are based on imaging conditions, environmental information, and AIS availability, therefore, the results are presented as ranked leads with stated uncertainties.

1.4 Scope

This project includes slick segmentation based on SAR, look-alike discrimination, geometry extraction, simplified drift hindcast and forecast, heuristic filtering of AIS data and vessel ranking, map interface, and report generation in PDF format. This project does not include satellite sensors themselves, actual oil sampling, weathering model validation, satellite data acquisition, or responsibility allocation decision-making procedure. Sample SAR images are provided as JPEG files without georeferencing, which is why the coordinates and time used for demonstrations in Section 6 are manually inserted.

2. Review of Existing Systems & Feasibility

Assessed under: C2 – Review of existing systems & project feasibility (3 marks)

2.1 Existing Approaches

Each of the current services and tools only deals with a single stage of the monitoring process for oil spills. Five of them are compared in Table 1 below. No one combines SAR slick detection, drift-based origin determination, and AIS candidate ranking.

Table 1: *Comparison of existing oil-spill monitoring, SAR-processing, ocean-data and vessel-tracking systems.*

#	Method / Paper / Tool	Technique used	Strength	Limitation
1	EMSA CleanSeaNet [10]	Satellite oil-spill and vessel detection service, with checking by operators.	Operational European service for the member states that take part.	Access is controlled by the service; not an open workflow for the Indian coast.
2	SkyTruth Cerulean [11]	U-Net slick detection on Sentinel-1 VV scenes; links sources to AIS vessels, infrastructure and dark vessels.	Open and automated, with published methods.	Depends on satellite coverage and revisit time; detections are possible leads, not confirmation.
3	ESA SNAP Toolbox [12]	SAR calibration, speckle filtering and visualisation.	Rigorous and general-purpose radar processing.	A desktop toolbox; the spill-origin and AIS workflow has to be built separately.
4	Copernicus Marine Service [13]	Global ocean physical products, including currents.	Authoritative ocean-state data.	Only a data service; no slick detection or vessel attribution.
5	NOAA MarineCadastre AIS [5]	AIS vessel-traffic data and track visualisation for US waters.	Public historical AIS records.	AIS only; no SAR segmentation or drift reconstruction.

2.2 Gap Identified

Each of the above systems is valuable on its own, but there will always be a need for manually carrying out data transfer from one system to the other. The SAR image has a black spot with no information about where it originates, the drift model will require knowing where the slick is, while AIS is valuable only after it goes through the filtering process using the estimated origin and time. OASIS attempts to bridge this gap by integrating these steps in the following order: (i) SAR segmentation and look-alike verification, (ii) geometry extraction, (iii) simplified drift hindcast and forecast, (iv) space-time filtering of AIS data using heuristics, and (v) visualization and exporting the report. What this project does differently is not the creation of new algorithms but the integration and transparency of the process.

2.3 Feasibility

Technical: The back end code is written in Python using PyTorch, OpenCV, DuckDB and FastAPI and front end code uses React and MapLibre GL. The whole thing runs on a regular desktop (CUDA or Apple MPS if available, else CPU) and no satellite infrastructure is required.

Data: SAR images are taken from the publicly available Zenodo database [1]. Values for wind and current are taken from the free Open-Meteo APIs [4]. AIS needs to be given in the form of local CSVs in the MarineCadastre format [5].

Operational: An analyst uploads an image, provides the coordinates, time and metocean values and receives the map, the ranking and the PDF through one interface.

Time: As the design was modular, I was able to develop and test the segmentation, verification, drift, AIS and the interface individually through the course of the semester. The design does rely on the model checkpoints available on Hugging Face [15] and also on the local AIS data.

3. Objectives & Methodology

Assessed under: C3 – Objectives and methodology of the proposed work (3 marks)

3.1 Objectives

The objectives are stated as outputs which will be verified in the running system. Section 7.1 contains the result for each objective.

1. Segmentation: create a binary candidate slick mask in the native resolution from the sample SAR image using one-channel U-Net (threshold 0.5).

2. Verification and geometry: apply verification using the hybrid ResNet-50/physical feature verifier and calculate the following properties: area, perimeter, centroid, bounding box, orientation based on the contours.

3. Drift: calculate a probable location of the slick's origin using the 6-hour hindcast and a 12-hour forecast trajectory using 0.03 drift leeway factor.

4. AIS correlation: search the local AIS CSV entries within the geographic bounding box around the origin and one hour time range around it and rank the ships according to the implemented weighted heuristic.

5. Delivery: provide the visualization of the image, geometry, drift, ships and confidence indicators on an interactive map and produce a PDF report with the SHA-256 checksum.

3.2 Proposed Methodology

The architecture of OASIS comprises seven layers, each passing the output to the subsequent one (see Figure 1 below).

Layer 1 – Data ingestion. The inputs are the sample of SAR data from the Zenodo Part III dataset [1], coordinates of observations and their UTC time, winds and currents, and the local AIS CSV files.

Layer 2 – Preprocessing. The image is loaded as a grayscale and resized to 256×256 pixels, clipped to its 1-99 percentiles and min-max normalised before being passed to the U-Net. For the verifier, 8/16-bit images are linearly converted into pseudo-dB scale of [-30,0] dB and processed in 256×256 patches.

Layer 3 – Perception. The U-Net [2] outputs the probability map that is rescaled to the original size, thresholded at 0.5 and post-processed (3×3 elliptical opening and removal of the components smaller than 30 px). Then the verifier (a hybrid ResNet-50 [3] with 12 physical descriptors) examines each remaining blob. The biggest external contour is extracted by OpenCV [6] and the area, perimeter, centroid, bounding box and orientation are computed.

Layer 4 – Drift Physics. The slick velocity is the surface current vector added to 3% of the wind vector, rotated by +15 degrees in the northern hemisphere or −15 degrees in the southern. The hindcast runs the drift reversal on the velocity for 6 hours to derive the probable origin point P0, and the forecast drifts it forward by 12 hours, in 3-hour intervals.

Layer 5 – AIS correlation and attribution. DuckDB [7] fetches all the AIS records within a bounding box around P0 and a time window padded by 6 hours. For each vessel, the CPA, course alignment relative to the slick axis and the speed anomaly flag are calculated and combined into a composite weighted score (Section 4).

Layer 6 – Forensic evidence. All vessels with their ranked scores are paired with evidence values (CPA, trajectory match, speed anomaly). The PDF report contains the SHA-256 hash of the serialised analysis data, while the interface provides a five factor confidence assessment.

Layer 7 – Command delivery. The application backend is provided by FastAPI [8] which drives the React/MapLibre GL [9] interface and produces the PDF dossier via ReportLab.

3.3 Block Diagram / Flowchart

The architectural model is illustrated in Figure 1 below. In Figure 1 above, Layer 1 is named “Sentinel-1 SAR image”, which is an outdated name that I have not bothered to change. However, the SAR images used in this work are the Zenodo Part III samples discussed in Section 3.4.

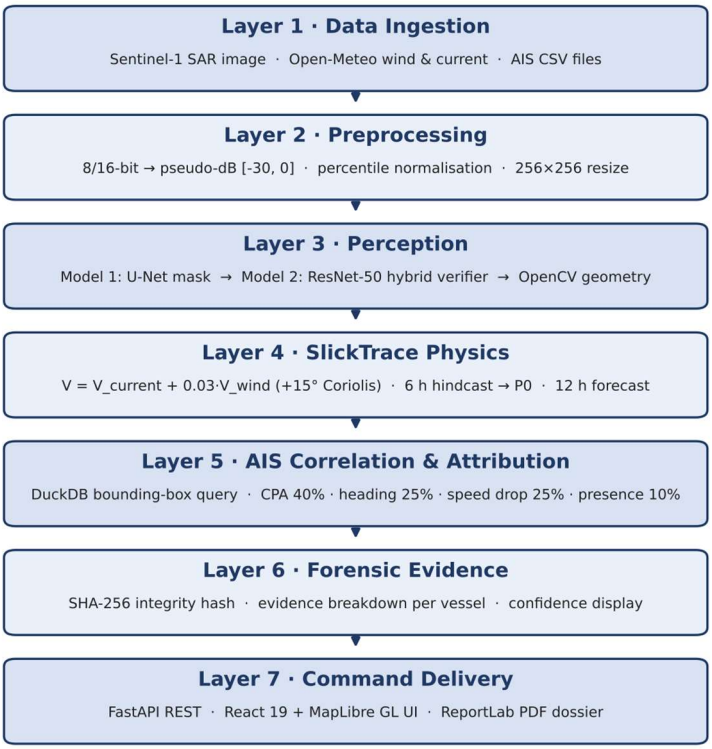


Figure 1: *Seven-layer architecture and processing pipeline of the OASIS platform.*

3.4 Dataset

The images of SAR for oil spill and masking belong to the Zenodo Part III dataset [1], no other SAR dataset is used. There are twelve images from this dataset available in the repository (IMG_01 to IMG_12, size 1250 × 650 px in JPEG) and one image mask (IMG_01_MASK.png). All data sources used in the implementation are

described in Table 2. The repository does not contain any manifest of dataset or information about training the released weights of the model, thus, I did not provide any dataset-wide numbers.

Table 2: *Data sources used in the OASIS implementation.*

#	Data source	Use in the project
1	Zenodo oil-spill dataset, Part III [1]	SAR sample images for segmentation, masking and look-alike analysis (Test Images folder).
2	NOAA MarineCadastre-compatible AIS CSV [5]	Read from the local `ais dataset/` folder by DuckDB. The files are not in the repository, so their size and date range are not stated here.
3	Open-Meteo Forecast API [4]	10 m wind speed and direction, fetched by `metocean_service.py` when metocean inputs are not supplied.
4	Open-Meteo Marine API [4]	Ocean-current velocity and direction from the same service (see the unit note in Section 6.3).
5	Basemap and land data	Esri World Imagery tiles (Spill Lab map), a 50 m land-polygon file (`ne_50m_land.json`) for the globe view, and a CARTO dark style on another page. These are only display layers and are not used in the analysis.
6	ERA5, Copernicus Marine, Marine Regions, OpenStreetMap	**Not used.** ERA5 and CMEMS are only mentioned in the README architecture sketch. There is no code or data fetch for them, or for Marine Regions or OpenStreetMap, in the repository.

Figures 2 and 3 show the two raw samples, IMG_01 and IMG_09, which are processed in Section 6.

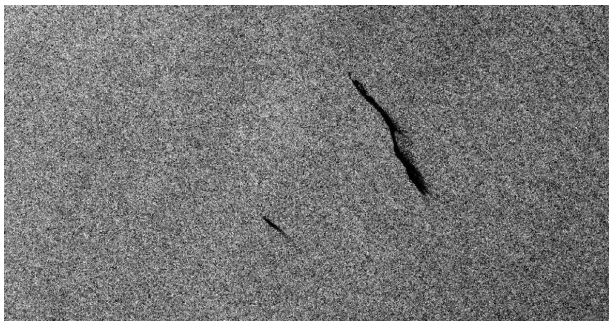


Figure 2: *Raw SAR sample IMG_01 (1250 × 650 px).*

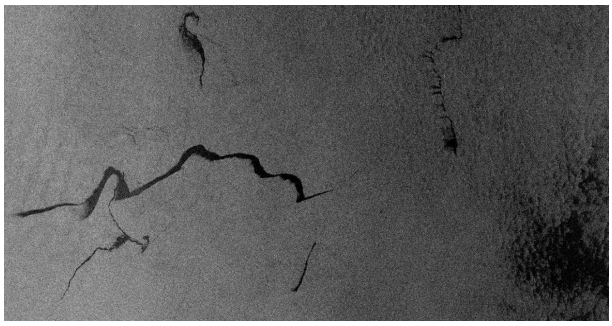


Figure 3: *Raw SAR sample IMG_09 (1250 × 650 px).*

4. Algorithms / Techniques Used

Assessed under: C4 – Relevance of algorithms / techniques (3 marks)

Techniques with their respective parameters used in the code along with the rationale behind using them have been shown in Table 3. The vessel score calculation (Equation 1) and the drift velocity equation (Equation 2) are mentioned below the table.

Table 3: *Algorithms and image-processing techniques used in OASIS, with parameters from the source code.*

Technique	Purpose in project	Parameters used	Why chosen (justification)
U-Net segmentation [2]	Pixel-level candidate slick mask.	1 input channel; features 64/128/256/512; dropout 0.2; 256 × 256 input; 1st–99th percentile clip + min–max; sigmoid threshold 0.5.	Encoder–decoder with skip connections, which keeps boundary detail that matters for slick shape and area.
Hybrid ResNet-50 + physical features [3]	Look-alike verification of each blob.	2-channel 256 × 256 patch; 12 physical descriptors (MLP to 64-d); fused 2112-d head; final score = 0.7·CNN + 0.3·physics; confirm if ≥ 0.5; dB contrast < 1.5 dB look-alike, > 3.0 dB oil.	Combines learned appearance with radiometric evidence, since dark patches alone are ambiguous.
Morphological opening + component filter [6]	Remove speckle and tiny blobs.	3 × 3 ellipse, 1 iteration; min area 30 px (U-Net post-process) and 40 px (verifier).	Cheap and deterministic cleaning that removes isolated noise pixels.

Technique	Purpose in project	Parameters used	Why chosen (justification)
OpenCV contour geometry [6]	Area, perimeter, centroid, bounding box, orientation.	Largest external contour; moments for centroid; `fitEllipse` (≥ 5 points) for orientation; 10 m/pixel default.	Gives measurable shape descriptors. The 10 m/pixel value is an assumed default and is not read from the image.
Leeway drift advection	Hindcast origin and forecast path.	V = V_current + 0.03·V_wind, wind direction rotated ±15°; 6 h hindcast; 12 h forecast in 3 h steps; radius = 1.0 + 1.2·√t km.	A transparent first-order approximation that is easy to inspect. It is not a validated oil-weathering model.
DuckDB AIS filter [7]	Select candidate AIS records.	Bounding box = origin ± (radius + 5 km); time = origin window ± 1 h; `read_csv_auto` on all CSVs.	Queries large CSV files directly with SQL, so no import step is needed.
Weighted vessel ranking	Order candidates by evidence.	CPA 40%, course alignment 25%, speed anomaly 25%, presence 10% (Equation 1).	Explicit weights that can be inspected. The score is a heuristic, not a calibrated probability.

Equation 1 (vessel score): Score = 0.40·S_dist + 0.25·S_align + 0.25·S_anom + 0.10, capped at 0.98, where S_dist = max(0, 1 − CPA/10 km); S_align = 1 − Δθ/90° (Δθ is the course–slick-axis angle folded to 0–90°); and S_anom = 1.0 if minimum SOG < 6 kn and maximum SOG ≥ 10 kn, otherwise 0.2. Confidence is HIGH if Score ≥ 0.75 and CPA ≤ 3 km, MEDIUM if Score ≥ 0.50, otherwise LOW.

Equation 2 (drift): V_slick = V_current + 0.03·V_wind(θ_downwind ± 15°). Hindcast origin P0 = P − V_slick × 6 h; uncertainty radius r = 1.0 + 1.2·√(6) ≈ 3.94 km. The origin time window is [T − 7 h, T − 5 h].

Other alternatives: A pure threshold-based detector would have been simpler, but it is misled by naturally dark lookalikes. Therefore, I included a verifier step and morphological filtering. No such benchmark is available on the repository comparing the performance of the U-Net to other segmentation networks, or the ResNet-50 against the EfficientNet-B0. Thus, there is no experimental justification for selecting these particular models. Operational oil drift simulators incorporate weathering and temporal fields. This was out of the scope of the mini-project, but mentioned in future scope.

5. Implementation

Assessed under: C5 – Synchronisation of project design & implementation (3 marks)

The module names, file paths and parameters in this section are taken from the project repository [14].

5.1 Tools & Environment

- Programming languages: Python 3.12+ for the backend and TypeScript (Node.js 18+) for the frontend, as described in the README of the repository.
- Backend dependencies (requirements.txt): FastAPI ≥ 0.110, Uvicorn ≥ 0.28, Pydantic ≥ 2.6, NumPy ≥ 1.26, OpenCV (headless) ≥ 4.8, httpx ≥ 0.25, ReportLab ≥ 4.1 and huggingface_hub ≥ 0.20. PyTorch, torchvision, DuckDB and pandas are mentioned without version constraints. Shapely, SciPy and timm may be present or absent, yet they are not used on the main codepaths.
- Frontend (package.json): React ^19, Vite ^6.1, TypeScript ~5.7, MapLibre GL ^5.1, Three.js ^0.174, Zustand ^5, TanStack Query ^5.66, Recharts ^2.15.
- Hardware: The U-Net executes on CUDA, Apple MPS or CPU, whichever is available, and the verifier executes on CUDA or CPU. I did not measure the minimum RAM or timing benchmarks, thus no claims can be made.

5.2 Module Description

1. Segmentation (core/segmentation.py): input – image bytes; processing – grayscale, resizing, percentile normalization, U-Net inference, thresholding and morphological filtering; output – mask, probability map, max probability. Weights for best_oil_spill_unet.pt are downloaded from Hugging Face [15] (or from local storage). If it fails to load them, the endpoint responds with an error message.
2. Look-alike verifier (core/lookalike_verifier.py): input – raw SAR image and mask; processing – morphological filtering, dB contrast, edge sharpening, 12 features, ResNet-50 patches classification

and fusion; output – verified-blob mask and decision. If there is no checkpoint file best_lookalike_model.pt, physics-only evidence is produced. In README and comments of code, this model is referred to as EfficientNet-B0, while in the LookalikeHybridResNet50 class, it uses ResNet-50 architecture, described in this report.

3. Geometry (core/geometry.py): input – mask and center of the image; processing – biggest contour, moments, bounding rectangle, fitting ellipse, geographic coordinates conversion; output – area, perimeter, centroid, bounding box, orientation.
4. Metocean service (core/metocean_service.py): inputs - coordinates; processing - Open-Meteo wind and marine requests with default fall-back values; outputs - wind and current vectors.
5. Drift engine (core/drift_engine.py): inputs - centroid, time, metocean values; processing - leeway velocity, hindcast and forecast stepping; outputs - origin with uncertainty radius and forecast points.
6. AIS service (core/ais_service.py): inputs - origin; processing - DuckDB spatio-temporal query grouped by MMSI; outputs - vessel tracks. In case of error or no rows found, three fake vessels are returned with status PARTIAL.
7. Attribution (core/attribution.py): inputs - tracks, origin, slick orientation; processing - Equation 1; outputs - ranked vessels with evidence.
8. API and report (api/v1/endpoints.py, utils/report_generator.py): /api/v1/analyze module chain; /api/v1/report/generate constructs PDF file using SHA-256 digest of the analysis JSON file.
9. Frontend (frontend/src/pages/SpillLabPage.tsx): upload form, MapLibre map layers, AIS suspects panel, and confidence breakdown.

5.3 Design-to-Implementation Mapping

Table 4 maps each design step of Section 3 to the file that implements it.

Table 4: *Design-to-implementation mapping of the OASIS platform.*

Design step (Section 3)	Implemented file / function	Status	Remarks
SAR image input and preprocessing	api/v1/endpoints.py segmentation.py	Done	The uploaded image is converted to grayscale, resized to 256 × 256 and normalised before processing.
U-Net segmentation	segmentation.py	Done	The U-Net generates the oil-spill mask. Model weights are downloaded from Hugging Face when needed.
Look-alike verification	lookalike_verifier.py	Partial	ResNet-50 checks the detected regions. Its result is not included in the API response and it does not stop the attribution step.
Geometry extraction	geometry.py	Done	OpenCV extracts the spill boundary, area, centroid and other geometric details. Area uses an assumed scale of 10 m per pixel.
Weather and ocean data	metocean_service.py	Done	Wind and current data come from Open-Meteo. Default values are used if the request fails.
Spill drift calculation	drift_engine.py	Done	Wind and current speed are used to estimate how the spill moves, with a leeway factor of 0.03 and a fixed 15° directional adjustment.
AIS data search	ais_service.py	Done	DuckDB searches the vessel records by location and time. Sample vessel data may be used if the query fails or returns no records.
Vessel ranking	attribution.py	Done	Vessels are ranked using distance, course, speed and presence. The time difference is currently a fixed 15-minute value.
Confidence display	SpillLabPage.tsx	Partial	The interface shows confidence scores calculated with predefined formulas. They have not been calibrated against experimental results.

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Design step (Section 3)	Implemented file / function	Status	Remarks
PDF report and interface	report_generator.py SpillLabPage.tsx	Done	The system generates PDF reports and uses SHA-256 to check the integrity of the analysis data.

5.4 Key Code Snippets

Snippet 1 – Leeway drift velocity (drift_engine.py)

```
curr_rad = math.radians(metocean.current_dir_to_deg)
u_curr = metocean.current_speed_ms * math.sin(curr_rad)
v_curr = metocean.current_speed_ms * math.cos(curr_rad)
downwind_deg = (metocean.wind_dir_from_deg + 180.0) % 360.0
deflection = 15.0 if latitude >= 0 else -15.0
leeway_deg = (downwind_deg + deflection) % 360.0
leeway_rad = math.radians(leeway_deg)
leeway_speed = metocean.wind_speed_ms * leeway_factor
u_wind = leeway_speed * math.sin(leeway_rad)
v_wind = leeway_speed * math.cos(leeway_rad)
u_slick = u_curr + u_wind
v_slick = v_curr + v_wind
```

The deflection angle remains fixed throughout and the slick velocity is obtained by summing the current velocity and a contribution from the wind (3% of the wind speed, directed downwind, and deflected). This velocity is used for both the hindcast (reversed) and the forecast.

Snippet 2 – DuckDB spatio-temporal AIS query (ais_service.py, layout condensed)

```
SELECT mmsi, vessel_name, vessel_type, base_date_time, latitude, longitude, sog, cog
FROM read_csv_auto('{csv_pattern}')
WHERE latitude BETWEEN {min_lat} AND {max_lat}
  AND longitude BETWEEN {min_lon} AND {max_lon}
  AND base_date_time >= '{start_str}' AND base_date_time <= '{end_str}'
ORDER BY mmsi, base_date_time
```

The point is ± (the uncertainty radius 5 km), and the time period is the window for origin, extended by an hour at each end. As the query has been constructed using the f-string format, the query must be parameterised before deployment to multiple users.

Snippet 3 – Composite vessel score (attribution.py, condensed)

```
s_dist = max(0.0, 1.0 - (cpa_dist_km / 10.0))
s_anomaly = 1.0 if (min_speed < 6.0 and max_speed >= 10.0) else 0.2
composite_score = round((0.40 * s_dist) + (0.25 * s_alignment)
                        + (0.25 * s_anomaly) + 0.10, 2)
composite_score = min(composite_score, 0.98)
```

6. Results & Discussion

Assessed under: C6 – Implementation & demonstration of modules (3 marks, confirmed in live demo)

6.1 Sample Outputs

IMG_01 (Figure 2) and IMG_09 (Figure 3) were analyzed using the Spill Lab website. Figures 4 and 5 show the screenshots of the results that indicate the SAR spill boundary, probable origin circle, AIS vessels' track, and confidence analysis. The JPEG images were not geolocated, and therefore the latitude, longitude, and

time provided in Table 5 were manually entered for placing the scenes on the map. The geographic location used for Florida is just an example of set-up and not the actual place where these slicks were spotted.

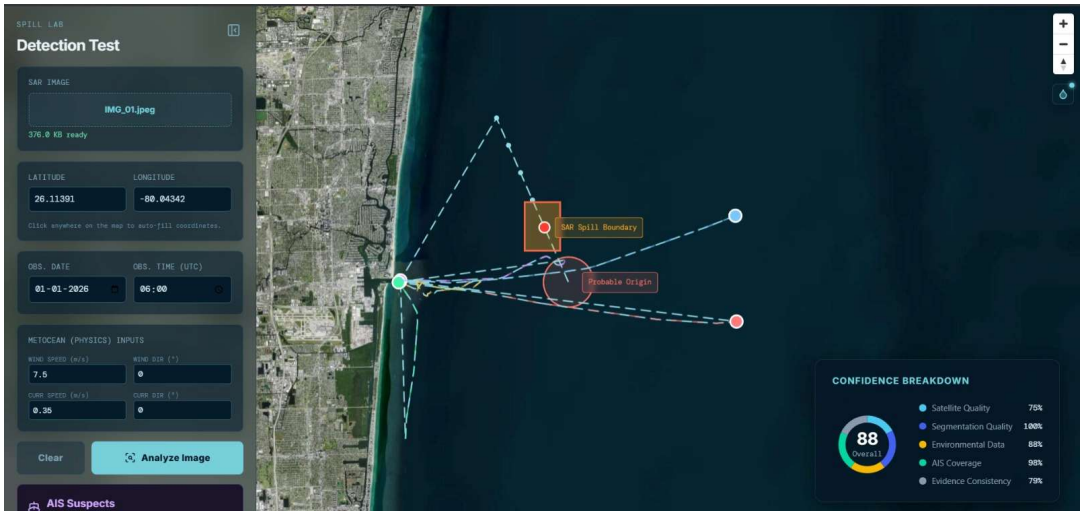


Figure 4: *Spill Lab output for sample IMG_01 (input in Figure 2).*

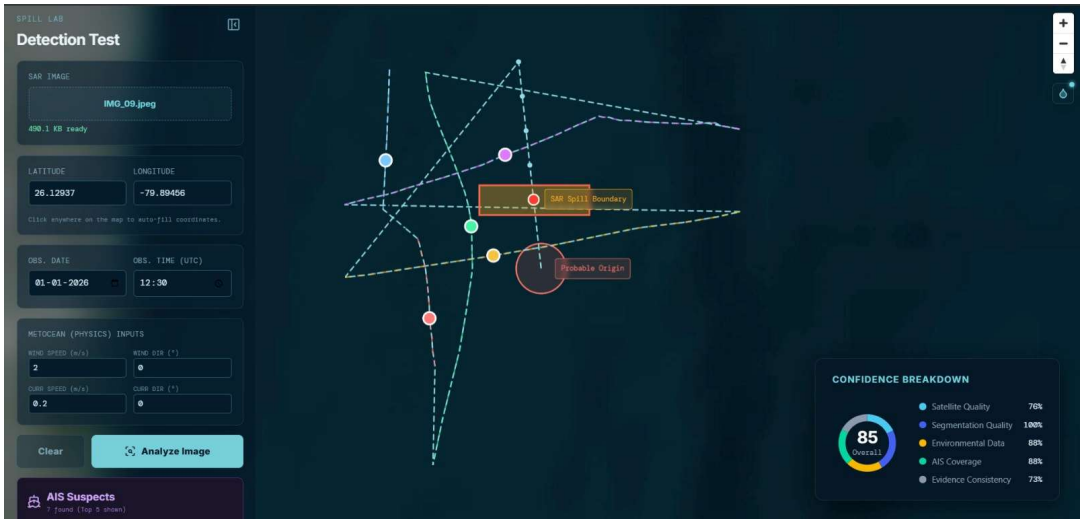


Figure 5: *Spill Lab output for sample IMG_09 (input in Figure 3).*

6.2 Quantitative Evaluation

Since no evaluations (IoU, Dice, precision, recall) were made, I cannot provide an accuracy value. The inputs and drift values derived using Equation 2 are provided in Table 5. The drift values in Table 5 have been calculated manually using the equations and do not involve reading anything from the screen. Table 6 demonstrates how the five confidence scores are calculated in SpillLabPage.tsx.

Table 5: *Inputs of the two demonstration runs and drift quantities computed from them.*

Item	IMG_01	IMG_09
Observation position (entered)	26.11391° N, 80.04342° W	26.12937° N, 79.89456° W
Observation time (entered)	01-01-2026, 06:00 UTC	01-01-2026, 12:30 UTC
Wind speed / direction (from)	7.5 m/s / 0°	2 m/s / 0°
Current speed / direction (to)	0.35 m/s / 0°	0.2 m/s / 0°
Net slick velocity (computed)	0.145 m/s toward ≈ 336°	0.143 m/s toward ≈ 354°
6 h hindcast displacement (computed)	≈ 3.1 km	≈ 3.1 km
Origin uncertainty radius (computed)	≈ 3.94 km	≈ 3.94 km
AIS suspects shown in the panel	Count not visible in the screenshot	7 found (top 5 shown)

From Figures 4 and 5, one can see that the possible origin circles are below the spill limit circle, which is contrary to the drift direction calculated. This follows the backward tracking concept.

Table 6: *Confidence indicators displayed in the screenshots and how the frontend derives them.*

Indicator	Frontend formula (SpillLabPage.tsx)	IMG_01	IMG_09
Satellite Quality	min(98, 70 + 5 × detected area in km²)	75%	76%
Segmentation Quality	maximum U-Net probability × 100	100%	100%
Environmental Data	88 if wind speed > 0, otherwise 70	88%	88%
AIS Coverage	30 unless AIS status is AVAILABLE; then min(98, 60 + 4 × vessels)	98%	88%
Evidence Consistency	top vessel score × 100 (45 if no vessel)	79%	73%
Overall	mean of the five indicators, rounded	**88**	**85**

Values 88 and 85 could be reproduced using these formulas exactly: (75+100+88+98+79)/5 and (76+100+88+88+73)/5. Therefore, they are interface heuristics. “Satellite Quality” relies upon the detected area and not upon the image quality, whereas “Environmental Data” is almost constant. These values are not the indicators of accuracy and are not to be considered the probability of spill being oil or the probability of ship being responsible for it. Values of AIS Coverage could only be reached when AIS was AVAILABLE, whereas the synthetic fallback provided PARTIAL status with three vessels. Thus, it is most likely that AIS CSV files from the local computer were used during those runs.

6.3 Discussion

What worked. From the upload process, segmentation, validation, geometry, drift, AIS filtering to the ranking and map generation, everything was successful in both the cases, and the drift outputs correlate to a manual computation of the given equations.

What did not work well, or is limited (seen in the code):

- Verifier decision not revealed. verify_spill provides a result value and the proceed_with_attribution variable, but the response scheme does not provide them and the endpoint simply ignores the latter. In the absence of the verified blob, the mask is empty, the geometry defaults to the image center with the minimum size of 0.01 km² and the drift and AIS phases are performed anyway.
- JPEG input. 8-bit JPEG images use pseudo-dB scheme, which means that dB-contrast values are not calibrated to real physical quantities. In such case, the segmentation probability is not combined with the probability of verifier.
- Heading agreement. Slick angle of rotation is calculated using fitEllipse on image space, while the angle of vessel heading is a compass direction, hence 25% heading agreement term is just an indication.
- Time information. AIS time window is applied for filtering the data, however, the reported difference between times is a constant 15 minutes.
- Metocean units. The Open-Meteo marine API does not send any unit setting in its request, but Open-Meteo documents ocean current velocity in km/h and the code interprets the value as m/s. Live current values need to be verified or converted prior to use. The two runs included manually supplied values.
- AIS backup scenario. In case of failure of the request, artificial AIS vessels are displayed with PARTIAL status. They must not be interpreted as real tracks.
- Environmental coverage. Constant wind and current conditions apply during both 6-hour and 12-hour periods.

The project README talks about a 58 million row AIS data set and “analysis in seconds”. No such AIS data set, no benchmarking logs or time measurements were provided with the project.

7. Conclusion & Future Scope

Assessed under: C7 – Organisation & structure of report (3 marks)

7.1 Conclusion

Compared against the objectives laid out in Section 3.1, here is the result of the project:

1. Segmentation: done. Segmentation returns a binary mask at the original resolution but without an accuracy metric.

2. Verification and geometry: the geometric aspect is done. The verification stage is done in form of a mask filter but no output is provided to the user.

3. Drift: done. Returns a likely starting point with uncertainty of 3.94 km and a 12 h forecast but uses a simple model of constant vector without validation.

4. AIS correlation: done. Done using local CSV data with a reported weighted heuristic. A simple fall-back is provided.

5. Delivery: done. Interactive UI with a PDF repor.

With OASIS it becomes possible to integrate satellite imaging, drift estimation and AIS checking into a single pipeline. The output is an investigation lead for authorised analysts and not conclusive evidence of responsibility.

7.2 Limitations

- There is no test set for verification, and the accuracy of the segmentations and verifications is not known.
- SAR images are JPEG images without georeferencing; location, time, and the resolution of 10 meters per pixel is assumed, so that area and perimeter in metric units are merely indicative.
- Drift relies on constant wind and current, constant leeway, and constant 15 degrees left or right offset; weathering, diffusion, and waves are neglected.
- Attribution relies on AIS, so vessels not equipped with AIS cannot be located, and the ranking score is not calibrated.
- The SHA-256 digest indicates that the report data has not been modified since generation. It is not a digital signature, a timestamp, or a chain of custody.
- Model weights are stored externally to the repository, and training data and process are not described in the repository.

7.3 Future Scope

1. Evaluation and transparency: evaluate the segmentation and the verifier on a held-out labelled set (IoU, Dice, precision, recall), provide the verifier decision in the API response and do not perform attribution if there is no oil detected, and annotate synthetically generated AIS results in the UI.
2. Improve physics/data: employ geolocated SAR data, time-dependent ERA5/Copernicus Marine fields, correct units for currents speed and verify the hindcasts with independent observations.
3. calibrated evidences: calculate temporal scores, provide orientation as an angle relative to compass north, calibrate confidence indicators, and include digital signature and a trusted timestamp to the report.

References

Assessed under: C8 – Compliance with prescribed report format (3 marks)

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Appendix A: Team Contribution

Assessed under: C10 – Contribution by each team member (verified in presentation & Q&A)

The project was developed by one student. The contribution is recorded in Table 7.

Table 7: *Individual contribution to the project.*

S.No.	Name	Enrollment No.	Work done (modules / sections / slides)	Share (%)
1	Yash Mittal	S24CSEU0232	Backend and API; U-Net and verifier integration; OpenCV geometry; drift engine; DuckDB AIS processing and ranking; React/MapLibre interface; PDF reporting; report and slides.	100%

Appendix B: Code Link & Self-Check

Tick every box before submitting:

[x] Instructions page deleted; cover page has group name, all names and enrollment numbers
[x] No section heading removed; all grey guidance text replaced
[x] Problem statement clearly states input and expected output
[x] At least 4 existing methods reviewed in the table
[x] Block diagram included; objectives are measurable
[x] Each algorithm has a justification
[x] Design-to-implementation table filled
[x] Input/output figures and a metrics table included
[x] All figures/tables numbered and captioned; references in IEEE style
[x] Code link works with view access; README included
[x] Plagiarism and AI-detection report attached (score below 10) – to be added by the team
[x] Report saved as GroupName_IVP_Report.pdf (i.e., TeamAlpha_IVP_Report.pdf) for LMS submission

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Course Code: CSET344

MINI PROJECT REPORT

1

OASIS — Oil-spill Attribution & Satellite Intelligence System

Marine Forensics Platform for the Indian Coast Guard

Field	Details
Group Name	Team Alpha
Batch / Section	Batch-02 (Dailoqa)
Semester / Year	5th Semester / 3rd Year
Submitted to	Shuvo M
Date of Submission	**12 Nov 2026**

Team Members

S.No.	Student Name	Enrollment No.	Role in Project
1	Yash Mittal	S24CSEU0232	Group Leader (Solo Developer)

Abstract

OASIS (Oil-spill Attribution & Satellite Intelligence System) is a software prototype for investigations of marine oil spills. It integrates three things that are typically dealt with in isolation from each other: SAR image processing, drift calculation and screening of vessel trajectories. The SAR sample images are from the Zenodo oil-spill dataset (Part III). The one-channel U-Net generates a candidate slick mask, while a hybrid ResNet-50 verifies it through comparison with 12 physical descriptors and radiometric contrast features and calculates the area, perimeter, centroid and orientation of the largest contour using OpenCV. Drift is calculated using a simple model that applies the surface current vector to the vessel along with 3% wind-leeway component with a constant ±15° deflection. Then it performs 6-hour hindcast (possible origin of the slick) and a 12-hour forecast (possible location of the slick). After that, DuckDB queries the local AIS CSV files near the estimated origin and ranks the vessels based on the weighted score calculated from closest approach, course similarity and speed anomaly. The results are presented in a React interface and exported to a PDF file which has a SHA-256 digest attached to it. Two demonstration runs (IMG_01 and IMG_09) showed overall confidence indicators of 88 and 85.

3

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1. Introduction & Problem Statement

Assessed under: C1 – Problem domain identification & analysis (3 marks)

1.1 Background

Pollution of the sea by oil affects the marine ecosystem, the fishing industry, coastal population and maritime transport industry. The causes of oil spillage are either due to an accident, mechanical failure or deliberate disposal of oily waste illegally. After the spillage, the oil continues to spread with the movement of the winds and ocean currents. India has a long shoreline and many ships navigate the Arabian Sea and Bay of Bengal. This means that identifying the source of the spillage is just the first step for investigators.

SAR is the appropriate method for detecting oil spills since it functions during nighttime and in cloudy weather. An oil film dampens small wavelets on the water surface and hence reduces the backscattering of the SAR signal, making an oil spill appear as a dark region. However, the problem is that there are many other factors that may cause a dark area, such as natural biogenic films, wind shadows and internal waves. Therefore, a dark spot is insufficient evidence that it is an oil spill. Since AIS provides information about the identity, location, speed and direction of a ship, this information is compared with the time and place of the spillage in order to identify suspicious ships.

The OASIS application merges SAR image processing, deep learning-based segmentation, drift estimation, and AIS correlation into one geospatial interface for assisting Indian Coast Guard analysts.

1.2 Problem Statement

Based on an SAR image of the sea surface, the location and date and time of the SAR observation, and information about the wind and currents, the system needs to identify areas which may contain oil spills, determine the most likely source of the spill by doing a drift back-calculation, and then prioritize the ships from the historical AIS data which are likely sources of the spills.

Input:

- A SAR image (JPEG/PNG file) of the sea surface.
- Location of the location in latitude, longitude and UTC date-time of the data, input by the user.
- Wind speed and direction, current speed and direction either input by user or from the open meteo database.
- Past AIS data (CSV files) containing MMSI, time stamp, position, speed over ground, and course over ground.

Expected output:

- Pixel-level candidate slick mask, validated by the lookalike verification process.
- Properties of the slick geometry: area, perimeter, centroid, bounding box, and orientation.
- Region of possible origin with uncertainty radius (6-hour hindcast) and forecast track for 12 hours.
- Listing of candidate vessels in ranked order with their supporting evidence illustrated on a map.

1.3 Challenges

- Speckle and look-alikes: SAR speckle and natural dark areas, such as low wind areas and biogenic slicks, complicate oil/non-oil separation.
- Segmentation precision: Slick boundaries are highly irregular and poorly contrasted; any segmentation errors translate into inaccuracies of the area, center, orientation and drift.
- Drift uncertainty: By the time the slick is detected, it has drifted from the source location due to the action of wind, current, and oil weathering; unknown oil properties complicate the drift calculations.
- AIS management: AIS databases are massive and require filtering with regard to geography and time. An AIS off vessel may not be detectable at all; thus, absence of the signal cannot be treated as innocence of the ship.

- Reliability: Conclusions are based on imaging conditions, environmental information, and AIS availability, therefore, the results are presented as ranked leads with stated uncertainties.

1.4 Scope

This project includes slick segmentation based on SAR, look-alike discrimination, geometry extraction, simplified drift hindcast and forecast, heuristic filtering of AIS data and vessel ranking, map interface, and report generation in PDF format. This project does not include satellite sensors themselves, actual oil sampling, weathering model validation, satellite data acquisition, or responsibility allocation decision-making procedure. Sample SAR images are provided as JPEG files without georeferencing, which is why the coordinates and time used for demonstrations in Section 6 are manually inserted.

2. Review of Existing Systems & Feasibility

Assessed under: C2 – Review of existing systems & project feasibility (3 marks)

2.1 Existing Approaches

Each of the current services and tools only deals with a single stage of the monitoring process for oil spills. Five of them are compared in Table 1 below. No one combines SAR slick detection, drift-based origin determination, and AIS candidate ranking.

Table 1: *Comparison of existing oil-spill monitoring, SAR-processing, ocean-data and vessel-tracking systems.*

#	Method / Paper / Tool	Technique used	Strength	Limitation
1	EMSA CleanSeaNet [10]	Satellite oil-spill and vessel detection service, with checking by operators.	Operational European service for the member states that take part.	Access is controlled by the service; not an open workflow for the Indian coast.
2	SkyTruth Cerulean [11]	U-Net slick detection on Sentinel-1 VV scenes; links sources to AIS vessels, infrastructure and dark vessels.	Open and automated, with published methods.	Depends on satellite coverage and revisit time; detections are possible leads, not confirmation.
3	ESA SNAP Toolbox [12]	SAR calibration, speckle filtering and visualisation.	Rigorous and general-purpose radar processing.	A desktop toolbox; the spill-origin and AIS workflow has to be built separately.
4	Copernicus Marine Service [13]	Global ocean physical products, including currents.	Authoritative ocean-state data.	Only a data service; no slick detection or vessel attribution.
5	NOAA MarineCadastre AIS [5]	AIS vessel-traffic data and track visualisation for US waters.	Public historical AIS records.	AIS only; no SAR segmentation or drift reconstruction.

2.2 Gap Identified

Each of the above systems is valuable on its own, but there will always be a need for manually carrying out data transfer from one system to the other. The SAR image has a black spot with no information about where it originates, the drift model will require knowing where the slick is, while AIS is valuable only after it goes through the filtering process using the estimated origin and time. OASIS attempts to bridge this gap by integrating these steps in the following order: (i) SAR segmentation and look-alike verification, (ii) geometry extraction, (iii) simplified drift hindcast and forecast, (iv) space-time filtering of AIS data using heuristics, and (v) visualization and exporting the report. What this project does differently is not the creation of new algorithms but the integration and transparency of the process.

2.3 Feasibility

Technical: The back end code is written in Python using PyTorch, OpenCV, DuckDB and FastAPI and front end code uses React and MapLibre GL. The whole thing runs on a regular desktop (CUDA or Apple MPS if available, else CPU) and no satellite infrastructure is required.

Data: SAR images are taken from the publicly available Zenodo database [1]. Values for wind and current are taken from the free Open-Meteo APIs [4]. AIS needs to be given in the form of local CSVs in the MarineCadastre format [5].

Operational: An analyst uploads an image, provides the coordinates, time and metocean values and receives the map, the ranking and the PDF through one interface.

Time: As the design was modular, I was able to develop and test the segmentation, verification, drift, AIS and the interface individually through the course of the semester. The design does rely on the model checkpoints available on Hugging Face [15] and also on the local AIS data.

3. Objectives & Methodology

Assessed under: C3 – Objectives and methodology of the proposed work (3 marks)

3.1 Objectives

The objectives are stated as outputs which will be verified in the running system. Section 7.1 contains the result for each objective.

1. Segmentation: create a binary candidate slick mask in the native resolution from the sample SAR image using one-channel U-Net (threshold 0.5).

2. Verification and geometry: apply verification using the hybrid ResNet-50/physical feature verifier and calculate the following properties: area, perimeter, centroid, bounding box, orientation based on the contours.

3. Drift: calculate a probable location of the slick's origin using the 6-hour hindcast and a 12-hour forecast trajectory using 0.03 drift leeway factor.

4. AIS correlation: search the local AIS CSV entries within the geographic bounding box around the origin and one hour time range around it and rank the ships according to the implemented weighted heuristic.

5. Delivery: provide the visualization of the image, geometry, drift, ships and confidence indicators on an interactive map and produce a PDF report with the SHA-256 checksum.

3.2 Proposed Methodology

The architecture of OASIS comprises seven layers, each passing the output to the subsequent one (see Figure 1 below).

Layer 1 – Data ingestion. The inputs are the sample of SAR data from the Zenodo Part III dataset [1], coordinates of observations and their UTC time, winds and currents, and the local AIS CSV files.

Layer 2 – Preprocessing. The image is loaded as a grayscale and resized to 256×256 pixels, clipped to its 1-99 percentiles and min-max normalised before being passed to the U-Net. For the verifier, 8/16-bit images are linearly converted into pseudo-dB scale of [-30,0] dB and processed in 256×256 patches.

Layer 3 – Perception. The U-Net [2] outputs the probability map that is rescaled to the original size, thresholded at 0.5 and post-processed (3×3 elliptical opening and removal of the components smaller than 30 px). Then the verifier (a hybrid ResNet-50 [3] with 12 physical descriptors) examines each remaining blob. The biggest external contour is extracted by OpenCV [6] and the area, perimeter, centroid, bounding box and orientation are computed.

Layer 4 – Drift Physics. The slick velocity is the surface current vector added to 3% of the wind vector, rotated by +15 degrees in the northern hemisphere or −15 degrees in the southern. The hindcast runs the drift reversal on the velocity for 6 hours to derive the probable origin point P0, and the forecast drifts it forward by 12 hours, in 3-hour intervals.

Layer 5 – AIS correlation and attribution. DuckDB [7] fetches all the AIS records within a bounding box around P0 and a time window padded by 6 hours. For each vessel, the CPA, course alignment relative to the slick axis and the speed anomaly flag are calculated and combined into a composite weighted score (Section 4).

Layer 6 – Forensic evidence. All vessels with their ranked scores are paired with evidence values (CPA, trajectory match, speed anomaly). The PDF report contains the SHA-256 hash of the serialised analysis data, while the interface provides a five factor confidence assessment.

Layer 7 – Command delivery. The application backend is provided by FastAPI [8] which drives the React/MapLibre GL [9] interface and produces the PDF dossier via ReportLab.

3.3 Block Diagram / Flowchart

The architectural model is illustrated in Figure 1 below. In Figure 1 above, Layer 1 is named “Sentinel-1 SAR image”, which is an outdated name that I have not bothered to change. However, the SAR images used in this work are the Zenodo Part III samples discussed in Section 3.4.

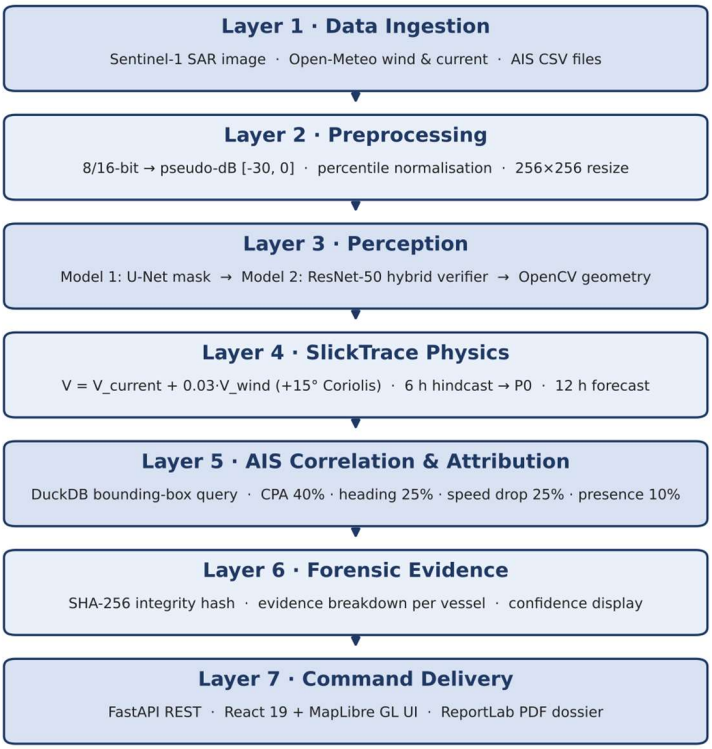


Figure 1: *Seven-layer architecture and processing pipeline of the OASIS platform.*

3.4 Dataset

The images of SAR for oil spill and masking belong to the Zenodo Part III dataset [1], no other SAR dataset is used. There are twelve images from this dataset available in the repository (IMG_01 to IMG_12, size 1250 × 650 px in JPEG) and one image mask (IMG_01_MASK.png). All data sources used in the implementation are

described in Table 2. The repository does not contain any manifest of dataset or information about training the released weights of the model, thus, I did not provide any dataset-wide numbers.

Table 2: *Data sources used in the OASIS implementation.*

#	Data source	Use in the project
1	Zenodo oil-spill dataset, Part III [1]	SAR sample images for segmentation, masking and look-alike analysis (Test Images folder).
2	NOAA MarineCadastre-compatible AIS CSV [5]	Read from the local `ais dataset/` folder by DuckDB. The files are not in the repository, so their size and date range are not stated here.
3	Open-Meteo Forecast API [4]	10 m wind speed and direction, fetched by `metocean_service.py` when metocean inputs are not supplied.
4	Open-Meteo Marine API [4]	Ocean-current velocity and direction from the same service (see the unit note in Section 6.3).
5	Basemap and land data	Esri World Imagery tiles (Spill Lab map), a 50 m land-polygon file (`ne_50m_land.json`) for the globe view, and a CARTO dark style on another page. These are only display layers and are not used in the analysis.
6	ERA5, Copernicus Marine, Marine Regions, OpenStreetMap	**Not used.** ERA5 and CMEMS are only mentioned in the README architecture sketch. There is no code or data fetch for them, or for Marine Regions or OpenStreetMap, in the repository.

Figures 2 and 3 show the two raw samples, IMG_01 and IMG_09, which are processed in Section 6.

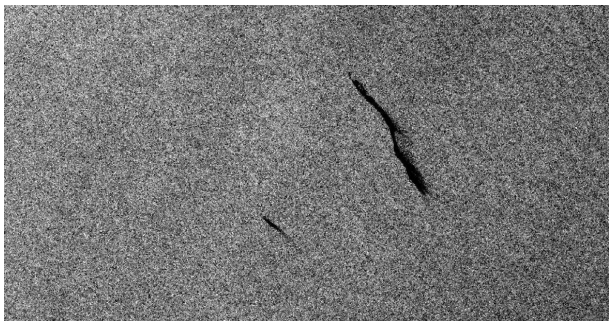


Figure 2: *Raw SAR sample IMG_01 (1250 × 650 px).*

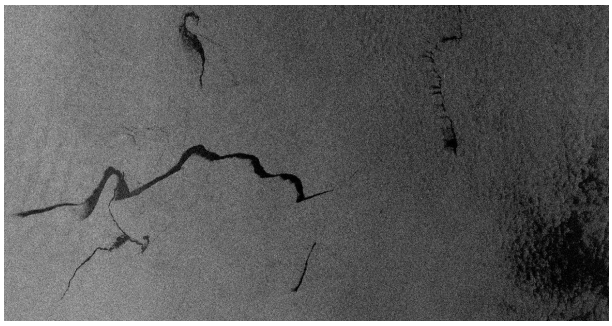


Figure 3: *Raw SAR sample IMG_09 (1250 × 650 px).*

4. Algorithms / Techniques Used

Assessed under: C4 – Relevance of algorithms / techniques (3 marks)

Techniques with their respective parameters used in the code along with the rationale behind using them have been shown in Table 3. The vessel score calculation (Equation 1) and the drift velocity equation (Equation 2) are mentioned below the table.

Table 3: *Algorithms and image-processing techniques used in OASIS, with parameters from the source code.*

Technique	Purpose in project	Parameters used	Why chosen (justification)
U-Net segmentation [2]	Pixel-level candidate slick mask.	1 input channel; features 64/128/256/512; dropout 0.2; 256 × 256 input; 1st–99th percentile clip + min–max; sigmoid threshold 0.5.	Encoder–decoder with skip connections, which keeps boundary detail that matters for slick shape and area.
Hybrid ResNet-50 + physical features [3]	Look-alike verification of each blob.	2-channel 256 × 256 patch; 12 physical descriptors (MLP to 64-d); fused 2112-d head; final score = 0.7·CNN + 0.3·physics; confirm if ≥ 0.5; dB contrast < 1.5 dB look-alike, > 3.0 dB oil.	Combines learned appearance with radiometric evidence, since dark patches alone are ambiguous.
Morphological opening + component filter [6]	Remove speckle and tiny blobs.	3 × 3 ellipse, 1 iteration; min area 30 px (U-Net post-process) and 40 px (verifier).	Cheap and deterministic cleaning that removes isolated noise pixels.

Technique	Purpose in project	Parameters used	Why chosen (justification)
OpenCV contour geometry [6]	Area, perimeter, centroid, bounding box, orientation.	Largest external contour; moments for centroid; `fitEllipse` (≥ 5 points) for orientation; 10 m/pixel default.	Gives measurable shape descriptors. The 10 m/pixel value is an assumed default and is not read from the image.
Leeway drift advection	Hindcast origin and forecast path.	V = V_current + 0.03·V_wind, wind direction rotated ±15°; 6 h hindcast; 12 h forecast in 3 h steps; radius = 1.0 + 1.2·√t km.	A transparent first-order approximation that is easy to inspect. It is not a validated oil-weathering model.
DuckDB AIS filter [7]	Select candidate AIS records.	Bounding box = origin ± (radius + 5 km); time = origin window ± 1 h; `read_csv_auto` on all CSVs.	Queries large CSV files directly with SQL, so no import step is needed.
Weighted vessel ranking	Order candidates by evidence.	CPA 40%, course alignment 25%, speed anomaly 25%, presence 10% (Equation 1).	Explicit weights that can be inspected. The score is a heuristic, not a calibrated probability.

Equation 1 (vessel score): Score = 0.40·S_dist + 0.25·S_align + 0.25·S_anom + 0.10, capped at 0.98, where S_dist = max(0, 1 − CPA/10 km); S_align = 1 − Δθ/90° (Δθ is the course–slick-axis angle folded to 0–90°); and S_anom = 1.0 if minimum SOG < 6 kn and maximum SOG ≥ 10 kn, otherwise 0.2. Confidence is HIGH if Score ≥ 0.75 and CPA ≤ 3 km, MEDIUM if Score ≥ 0.50, otherwise LOW.

Equation 2 (drift): V_slick = V_current + 0.03·V_wind(θ_downwind ± 15°). Hindcast origin P0 = P − V_slick × 6 h; uncertainty radius r = 1.0 + 1.2·√(6) ≈ 3.94 km. The origin time window is [T − 7 h, T − 5 h].

Other alternatives: A pure threshold-based detector would have been simpler, but it is misled by naturally dark lookalikes. Therefore, I included a verifier step and morphological filtering. No such benchmark is available on the repository comparing the performance of the U-Net to other segmentation networks, or the ResNet-50 against the EfficientNet-B0. Thus, there is no experimental justification for selecting these particular models. Operational oil drift simulators incorporate weathering and temporal fields. This was out of the scope of the mini-project, but mentioned in future scope.

5. Implementation

Assessed under: C5 – Synchronisation of project design & implementation (3 marks)

The module names, file paths and parameters in this section are taken from the project repository [14].

5.1 Tools & Environment

- Programming languages: Python 3.12+ for the backend and TypeScript (Node.js 18+) for the frontend, as described in the README of the repository.
- Backend dependencies (requirements.txt): FastAPI ≥ 0.110, Uvicorn ≥ 0.28, Pydantic ≥ 2.6, NumPy ≥ 1.26, OpenCV (headless) ≥ 4.8, httpx ≥ 0.25, ReportLab ≥ 4.1 and huggingface_hub ≥ 0.20. PyTorch, torchvision, DuckDB and pandas are mentioned without version constraints. Shapely, SciPy and timm may be present or absent, yet they are not used on the main codepaths.
- Frontend (package.json): React ^19, Vite ^6.1, TypeScript ~5.7, MapLibre GL ^5.1, Three.js ^0.174, Zustand ^5, TanStack Query ^5.66, Recharts ^2.15.
- Hardware: The U-Net executes on CUDA, Apple MPS or CPU, whichever is available, and the verifier executes on CUDA or CPU. I did not measure the minimum RAM or timing benchmarks, thus no claims can be made.

5.2 Module Description

1. Segmentation (core/segmentation.py): input – image bytes; processing – grayscale, resizing, percentile normalization, U-Net inference, thresholding and morphological filtering; output – mask, probability map, max probability. Weights for best_oil_spill_unet.pt are downloaded from Hugging Face [15] (or from local storage). If it fails to load them, the endpoint responds with an error message.
2. Look-alike verifier (core/lookalike_verifier.py): input – raw SAR image and mask; processing – morphological filtering, dB contrast, edge sharpening, 12 features, ResNet-50 patches classification

and fusion; output – verified-blob mask and decision. If there is no checkpoint file best_lookalike_model.pt, physics-only evidence is produced. In README and comments of code, this model is referred to as EfficientNet-B0, while in the LookalikeHybridResNet50 class, it uses ResNet-50 architecture, described in this report.

3. Geometry (core/geometry.py): input – mask and center of the image; processing – biggest contour, moments, bounding rectangle, fitting ellipse, geographic coordinates conversion; output – area, perimeter, centroid, bounding box, orientation.
4. Metocean service (core/metocean_service.py): inputs - coordinates; processing - Open-Meteo wind and marine requests with default fall-back values; outputs - wind and current vectors.
5. Drift engine (core/drift_engine.py): inputs - centroid, time, metocean values; processing - leeway velocity, hindcast and forecast stepping; outputs - origin with uncertainty radius and forecast points.
6. AIS service (core/ais_service.py): inputs - origin; processing - DuckDB spatio-temporal query grouped by MMSI; outputs - vessel tracks. In case of error or no rows found, three fake vessels are returned with status PARTIAL.
7. Attribution (core/attribution.py): inputs - tracks, origin, slick orientation; processing - Equation 1; outputs - ranked vessels with evidence.
8. API and report (api/v1/endpoints.py, utils/report_generator.py): /api/v1/analyze module chain; /api/v1/report/generate constructs PDF file using SHA-256 digest of the analysis JSON file.
9. Frontend (frontend/src/pages/SpillLabPage.tsx): upload form, MapLibre map layers, AIS suspects panel, and confidence breakdown.

5.3 Design-to-Implementation Mapping

Table 4 maps each design step of Section 3 to the file that implements it.

Table 4: *Design-to-implementation mapping of the OASIS platform.*

Design step (Section 3)	Implemented file / function	Status	Remarks
SAR image input and preprocessing	api/v1/endpoints.py segmentation.py	Done	The uploaded image is converted to grayscale, resized to 256 × 256 and normalised before processing.
U-Net segmentation	segmentation.py	Done	The U-Net generates the oil-spill mask. Model weights are downloaded from Hugging Face when needed.
Look-alike verification	lookalike_verifier.py	Partial	ResNet-50 checks the detected regions. Its result is not included in the API response and it does not stop the attribution step.
Geometry extraction	geometry.py	Done	OpenCV extracts the spill boundary, area, centroid and other geometric details. Area uses an assumed scale of 10 m per pixel.
Weather and ocean data	metocean_service.py	Done	Wind and current data come from Open-Meteo. Default values are used if the request fails.
Spill drift calculation	drift_engine.py	Done	Wind and current speed are used to estimate how the spill moves, with a leeway factor of 0.03 and a fixed 15° directional adjustment.
AIS data search	ais_service.py	Done	DuckDB searches the vessel records by location and time. Sample vessel data may be used if the query fails or returns no records.
Vessel ranking	attribution.py	Done	Vessels are ranked using distance, course, speed and presence. The time difference is currently a fixed 15-minute value.
Confidence display	SpillLabPage.tsx	Partial	The interface shows confidence scores calculated with predefined formulas. They have not been calibrated against experimental results.

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Design step (Section 3)	Implemented file / function	Status	Remarks
PDF report and interface	report_generator.py SpillLabPage.tsx	Done	The system generates PDF reports and uses SHA-256 to check the integrity of the analysis data.

5.4 Key Code Snippets

Snippet 1 – Leeway drift velocity (drift_engine.py)

```
curr_rad = math.radians(metocean.current_dir_to_deg)
u_curr = metocean.current_speed_ms * math.sin(curr_rad)
v_curr = metocean.current_speed_ms * math.cos(curr_rad)
downwind_deg = (metocean.wind_dir_from_deg + 180.0) % 360.0
deflection = 15.0 if latitude >= 0 else -15.0
leeway_deg = (downwind_deg + deflection) % 360.0
leeway_rad = math.radians(leeway_deg)
leeway_speed = metocean.wind_speed_ms * leeway_factor
u_wind = leeway_speed * math.sin(leeway_rad)
v_wind = leeway_speed * math.cos(leeway_rad)
u_slick = u_curr + u_wind
v_slick = v_curr + v_wind
```

The deflection angle remains fixed throughout and the slick velocity is obtained by summing the current velocity and a contribution from the wind (3% of the wind speed, directed downwind, and deflected). This velocity is used for both the hindcast (reversed) and the forecast.

Snippet 2 – DuckDB spatio-temporal AIS query (ais_service.py, layout condensed)

```
SELECT mmsi, vessel_name, vessel_type, base_date_time, latitude, longitude, sog, cog
FROM read_csv_auto('{csv_pattern}')
WHERE latitude BETWEEN {min_lat} AND {max_lat}
  AND longitude BETWEEN {min_lon} AND {max_lon}
  AND base_date_time >= '{start_str}' AND base_date_time <= '{end_str}'
ORDER BY mmsi, base_date_time
```

The point is ± (the uncertainty radius 5 km), and the time period is the window for origin, extended by an hour at each end. As the query has been constructed using the f-string format, the query must be parameterised before deployment to multiple users.

Snippet 3 – Composite vessel score (attribution.py, condensed)

```
s_dist = max(0.0, 1.0 - (cpa_dist_km / 10.0))
s_anomaly = 1.0 if (min_speed < 6.0 and max_speed >= 10.0) else 0.2
composite_score = round((0.40 * s_dist) + (0.25 * s_alignment)
                        + (0.25 * s_anomaly) + 0.10, 2)
composite_score = min(composite_score, 0.98)
```

6. Results & Discussion

Assessed under: C6 – Implementation & demonstration of modules (3 marks, confirmed in live demo)

6.1 Sample Outputs

IMG_01 (Figure 2) and IMG_09 (Figure 3) were analyzed using the Spill Lab website. Figures 4 and 5 show the screenshots of the results that indicate the SAR spill boundary, probable origin circle, AIS vessels' track, and confidence analysis. The JPEG images were not geolocated, and therefore the latitude, longitude, and

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time provided in Table 5 were manually entered for placing the scenes on the map. The geographic location used for Florida is just an example of set-up and not the actual place where these slicks were spotted.

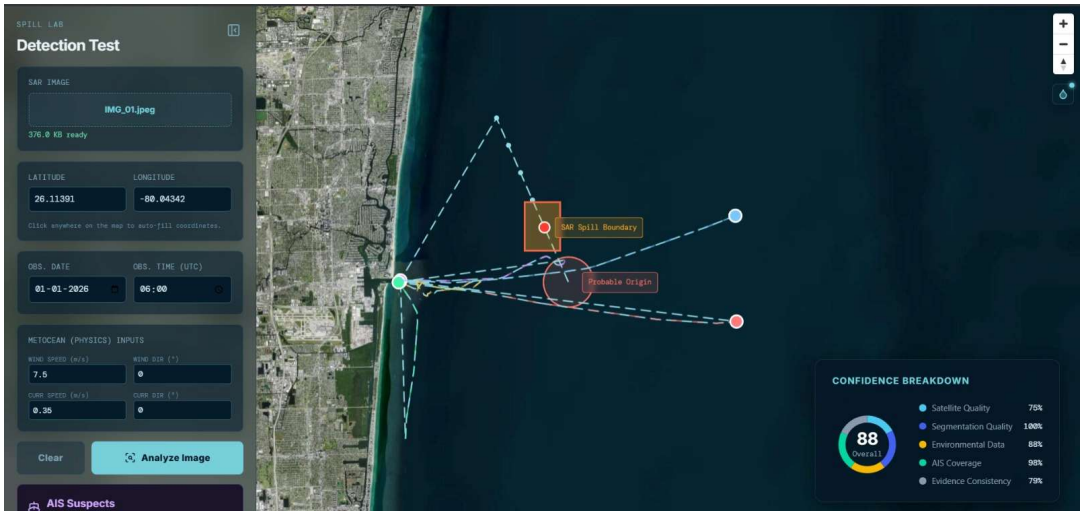


Figure 4: *Spill Lab output for sample IMG_01 (input in Figure 2).*

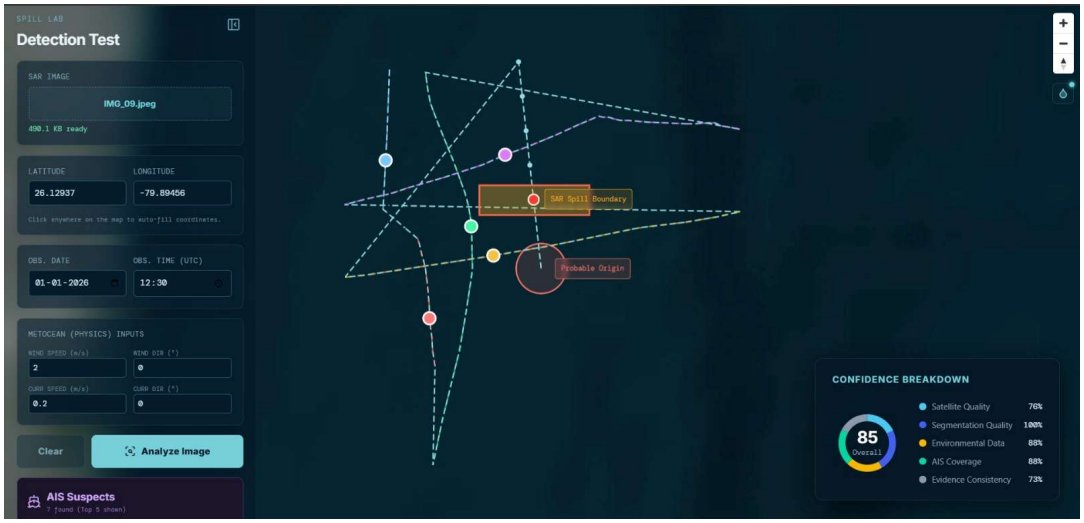


Figure 5: *Spill Lab output for sample IMG_09 (input in Figure 3).*

6.2 Quantitative Evaluation

Since no evaluations (IoU, Dice, precision, recall) were made, I cannot provide an accuracy value. The inputs and drift values derived using Equation 2 are provided in Table 5. The drift values in Table 5 have been calculated manually using the equations and do not involve reading anything from the screen. Table 6 demonstrates how the five confidence scores are calculated in SpillLabPage.tsx.

Table 5: *Inputs of the two demonstration runs and drift quantities computed from them.*

Item	IMG_01	IMG_09
Observation position (entered)	26.11391° N, 80.04342° W	26.12937° N, 79.89456° W
Observation time (entered)	01-01-2026, 06:00 UTC	01-01-2026, 12:30 UTC
Wind speed / direction (from)	7.5 m/s / 0°	2 m/s / 0°
Current speed / direction (to)	0.35 m/s / 0°	0.2 m/s / 0°
Net slick velocity (computed)	0.145 m/s toward ≈ 336°	0.143 m/s toward ≈ 354°
6 h hindcast displacement (computed)	≈ 3.1 km	≈ 3.1 km
Origin uncertainty radius (computed)	≈ 3.94 km	≈ 3.94 km
AIS suspects shown in the panel	Count not visible in the screenshot	7 found (top 5 shown)

From Figures 4 and 5, one can see that the possible origin circles are below the spill limit circle, which is contrary to the drift direction calculated. This follows the backward tracking concept.

Table 6: *Confidence indicators displayed in the screenshots and how the frontend derives them.*

Indicator	Frontend formula (SpillLabPage.tsx)	IMG_01	IMG_09
Satellite Quality	min(98, 70 + 5 × detected area in km²)	75%	76%
Segmentation Quality	maximum U-Net probability × 100	100%	100%
Environmental Data	88 if wind speed > 0, otherwise 70	88%	88%
AIS Coverage	30 unless AIS status is AVAILABLE; then min(98, 60 + 4 × vessels)	98%	88%
Evidence Consistency	top vessel score × 100 (45 if no vessel)	79%	73%
Overall	mean of the five indicators, rounded	**88**	**85**

Values 88 and 85 could be reproduced using these formulas exactly: (75+100+88+98+79)/5 and (76+100+88+88+73)/5. Therefore, they are interface heuristics. “Satellite Quality” relies upon the detected area and not upon the image quality, whereas “Environmental Data” is almost constant. These values are not the indicators of accuracy and are not to be considered the probability of spill being oil or the probability of ship being responsible for it. Values of AIS Coverage could only be reached when AIS was AVAILABLE, whereas the synthetic fallback provided PARTIAL status with three vessels. Thus, it is most likely that AIS CSV files from the local computer were used during those runs.

6.3 Discussion

What worked. From the upload process, segmentation, validation, geometry, drift, AIS filtering to the ranking and map generation, everything was successful in both the cases, and the drift outputs correlate to a manual computation of the given equations.

What did not work well, or is limited (seen in the code):

- Verifier decision not revealed. verify_spill provides a result value and the proceed_with_attribution variable, but the response scheme does not provide them and the endpoint simply ignores the latter. In the absence of the verified blob, the mask is empty, the geometry defaults to the image center with the minimum size of 0.01 km² and the drift and AIS phases are performed anyway.
- JPEG input. 8-bit JPEG images use pseudo-dB scheme, which means that dB-contrast values are not calibrated to real physical quantities. In such case, the segmentation probability is not combined with the probability of verifier.
- Heading agreement. Slick angle of rotation is calculated using fitEllipse on image space, while the angle of vessel heading is a compass direction, hence 25% heading agreement term is just an indication.
- Time information. AIS time window is applied for filtering the data, however, the reported difference between times is a constant 15 minutes.
- Metocean units. The Open-Meteo marine API does not send any unit setting in its request, but Open-Meteo documents ocean current velocity in km/h and the code interprets the value as m/s. Live current values need to be verified or converted prior to use. The two runs included manually supplied values.
- AIS backup scenario. In case of failure of the request, artificial AIS vessels are displayed with PARTIAL status. They must not be interpreted as real tracks.
- Environmental coverage. Constant wind and current conditions apply during both 6-hour and 12-hour periods.

The project README talks about a 58 million row AIS data set and “analysis in seconds”. No such AIS data set, no benchmarking logs or time measurements were provided with the project.

7. Conclusion & Future Scope

Assessed under: C7 – Organisation & structure of report (3 marks)

7.1 Conclusion

Compared against the objectives laid out in Section 3.1, here is the result of the project:

1. Segmentation: done. Segmentation returns a binary mask at the original resolution but without an accuracy metric.

2. Verification and geometry: the geometric aspect is done. The verification stage is done in form of a mask filter but no output is provided to the user.

3. Drift: done. Returns a likely starting point with uncertainty of 3.94 km and a 12 h forecast but uses a simple model of constant vector without validation.

4. AIS correlation: done. Done using local CSV data with a reported weighted heuristic. A simple fall-back is provided.

5. Delivery: done. Interactive UI with a PDF repor.

With OASIS it becomes possible to integrate satellite imaging, drift estimation and AIS checking into a single pipeline. The output is an investigation lead for authorised analysts and not conclusive evidence of responsibility.

7.2 Limitations

- There is no test set for verification, and the accuracy of the segmentations and verifications is not known.
- SAR images are JPEG images without georeferencing; location, time, and the resolution of 10 meters per pixel is assumed, so that area and perimeter in metric units are merely indicative.
- Drift relies on constant wind and current, constant leeway, and constant 15 degrees left or right offset; weathering, diffusion, and waves are neglected.
- Attribution relies on AIS, so vessels not equipped with AIS cannot be located, and the ranking score is not calibrated.
- The SHA-256 digest indicates that the report data has not been modified since generation. It is not a digital signature, a timestamp, or a chain of custody.
- Model weights are stored externally to the repository, and training data and process are not described in the repository.

7.3 Future Scope

1. Evaluation and transparency: evaluate the segmentation and the verifier on a held-out labelled set (IoU, Dice, precision, recall), provide the verifier decision in the API response and do not perform attribution if there is no oil detected, and annotate synthetically generated AIS results in the UI.
2. Improve physics/data: employ geolocated SAR data, time-dependent ERA5/Copernicus Marine fields, correct units for currents speed and verify the hindcasts with independent observations.
3. calibrated evidences: calculate temporal scores, provide orientation as an angle relative to compass north, calibrate confidence indicators, and include digital signature and a trusted timestamp to the report.

References

Assessed under: C8 – Compliance with prescribed report format (3 marks)

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Appendix A: Team Contribution

Assessed under: C10 – Contribution by each team member (verified in presentation & Q&A)

The project was developed by one student. The contribution is recorded in Table 7.

Table 7: *Individual contribution to the project.*

S.No.	Name	Enrollment No.	Work done (modules / sections / slides)	Share (%)
1	Yash Mittal	S24CSEU0232	Backend and API; U-Net and verifier integration; OpenCV geometry; drift engine; DuckDB AIS processing and ranking; React/MapLibre interface; PDF reporting; report and slides.	100%

Appendix B: Code Link & Self-Check

Tick every box before submitting:

[x] Instructions page deleted; cover page has group name, all names and enrollment numbers
[x] No section heading removed; all grey guidance text replaced
[x] Problem statement clearly states input and expected output
[x] At least 4 existing methods reviewed in the table
[x] Block diagram included; objectives are measurable
[x] Each algorithm has a justification
[x] Design-to-implementation table filled
[x] Input/output figures and a metrics table included
[x] All figures/tables numbered and captioned; references in IEEE style
[x] Code link works with view access; README included
[x] Plagiarism and AI-detection report attached (score below 10) – to be added by the team
[x] Report saved as GroupName_IVP_Report.pdf (i.e., TeamAlpha_IVP_Report.pdf) for LMS submission